# CLOVE: Continual Learning for Open-World Visual Evaluation in Agriculture

Tirtho Roy[1], Shreyan Ganguly[1], Asheesh K. Singh[1], Baskar Ganapathysubramanian[1], Chinmay Hegde[2], Arti Singh[1], Soumik Sarkar[1,†]

[1]Iowa State University, USA [2]New York University, USA [†]Corresponding author

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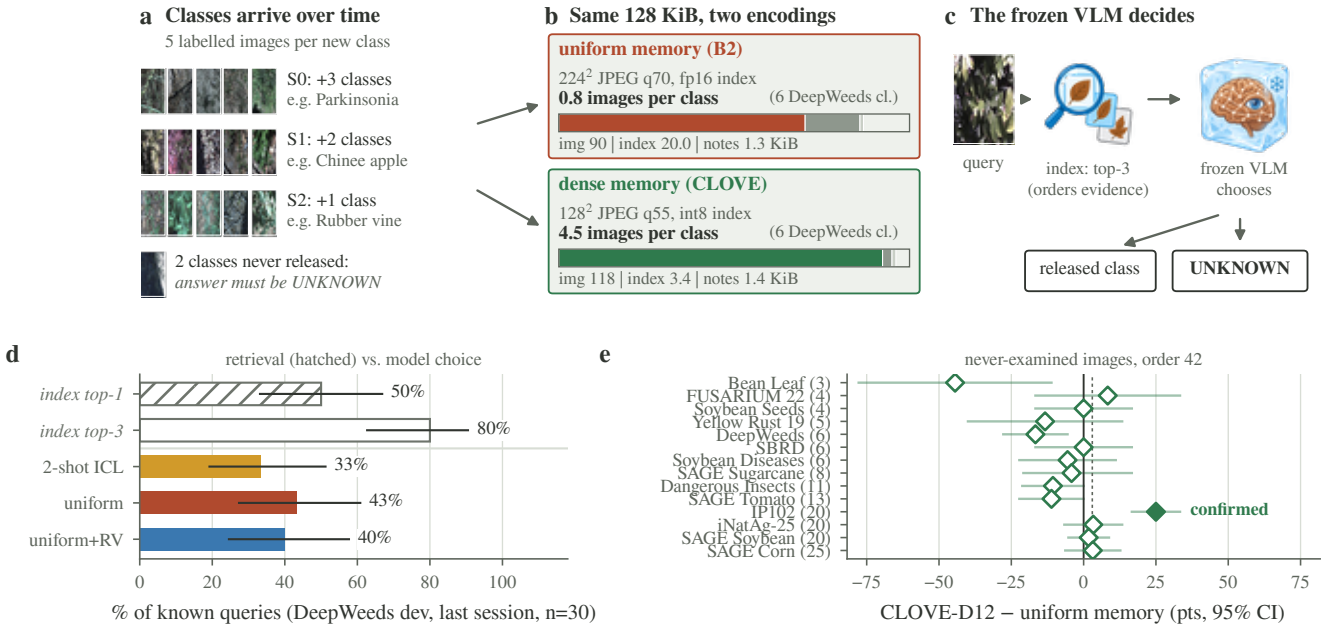



Figure 1. CLOVE at a glance. (a) Agricultural classes are released over sessions with five labelled images each (real DeepWeeds supports, class order 42); held-out classes are never released and must be answered UNKNOWN. (b) Both memories have the same 128 KiB budget. Bars show the measured bytes at the last session (images, index, notes); the denser encoding retains more references per class, and the realized count depends on the class inventory and image content (Fig. 4a). (c) The memory index only ranks and orders evidence; the frozen VLM chooses a released class or UNKNOWN. (d) Retrieval versus decision on DeepWeeds development data: the true class is in the index top three far more often than the model's forced choice is correct (Wilson 95% intervals). (e) Pre-registered confirmation on never-examined images (order 42): paired effect of CLOVE-D12 against the uniform memory with unadjusted 95% CI; filled marker: confirmed under the rule ($\geq$3 points, CI above zero, $p_{\text{Holm}} < 0.05$). Released classes in brackets. Photographs are real data; icons are illustrations.

## Abstract

CLOVE (Continual Learning for Open-world Visual Evaluation in agriculture) asks whether a small, locally runnable vision-language model can itself be the classifier when agricultural categories are released over time with five labelled images each, while images of never-taught categories keep arriving. Frozen Qwen2.5-VL-3B-Instruct [3] (arXiv 2025) chooses a released class or UNKNOWN; nothing is trained and only a 128 KiB memory changes. On identical supports, class orders and queries we find three things. (i) Generative collapse: as classes accumulate the model's decision degrades although retrieval covers the answer—on DeepWeeds development data the true class is in the index top three for 80% of images, but the forced choice is right for 43%. (ii) Agentic orchestration does not repair it: under pre-registered confirmation on never-examined image groups, exhaustive class verification is not confirmed and retrieve-then-verify is not con-

firmed. (iii) Memory optimisation helps where the byte budget is the bottleneck: a dense memory that stores several times more references per class, combined with retrieve-then-verify (CLOVE-D12), meets the pre-registered rule on 3 of 18 data sets (PlantDoc (+27 points vs B12); PlantVillage (+27 points vs B12); iNatAg-100 (+9 points vs B12)), is lower on 7 and otherwise indistinguishable. Frozen-feature readouts such as FeCAM [7] (NeurIPS 2023) and RanPAC [13] (NeurIPS 2023) remain more accurate in closed-set accuracy than any generative arm. All arms, including every failed one, are reported.

## 1. Introduction

A recognition system for crops, weeds and pests is updated as new categories become relevant; each update brings only a few labelled images, and images of categories that were never taught keep arriving. A frozen vision-language model (VLM) with an external memory is an attractive design for this setting: nothing is trained, the memory is inspectable, and the same model also answers free-form questions. Benchmarks such as AgEval [1] (WACV 2025) evaluate VLMs on fixed agricultural label sets, and agentic systems such as SAGE [2] (arXiv 2026) ground diagnoses in retrieved knowledge; neither releases classes over time under a memory budget, and neither lets a small local model act as the continual classifier.

We study exactly this setting and follow one line of evidence (Fig. 1). First, the model's own multiple-choice decision collapses as classes accumulate, although the memory index still ranks the true class near the top; predictions concentrate on a few attractive names and change with option order (Sec. 5.1). Second, the usual agentic remedies—SAGE-style inspection, retrieve-then-verify and exhaustive per-class verification—do not survive pre-registered confirmation (Sec. 5.2). Third, we optimise what the memory stores within the same byte budget, select the memory–orchestration combination on development data only, and confirm it on never-examined images of every eligible data set (Sec. 5.3). Fourth, we follow the result data set by data set (Sec. 5.4).

**Key finding**

The bottleneck of a frozen generative classifier is its decision among names, not visual coverage. Orchestration does not move it; storing more, smaller references per class does—but only where many classes compete for the budget.

Contributions. (1) A training-free, byte-bounded, open-world class-incremental protocol in which the VLM makes every decision, on 18 agricultural data sets, with a coverage audit of 21 candidates. (2) A diagnosis of generative collapse (coverage versus decision, attractors, order sensitivity). (3) Pre-registered negative results for agentic orchestration and verification. (4) A byte-optimised memory and a data-set-level account of when it helps. No component is claimed as new by itself.

## 2. Related Work

Agricultural VLM evaluation. AgEval [1] (WACV 2025) and AgroBench [18] (ICCV 2025) evaluate VLMs on static agricultural tasks; SAGE [2] (arXiv 2026) and AgriRAG [12] (CVPRW 2026) retrieve knowledge or images for a VLM. None releases categories over time or bounds memory.

Retrieval and in-context recognition. RAR [11] (arXiv 2024), Visual RAG [5] (arXiv 2025) and in-context classification with multimodal models [6, 9] (arXiv 2026, 2024) are the closest generic techniques; open-weight models gain little from many in-context examples [9].

Continual and open-world learning. FSCIL [19] (CVPR 2020), SLDA [8] (CVPRW 2020), FeCAM [7] (NeurIPS 2023), RanPAC [13] (NeurIPS 2023), TEEN [20] (NeurIPS 2023) and L2P [22] (CVPR 2022) learn readouts or prompts; InCA [15] (arXiv 2024) pairs a frozen language model with an external learner; open-world recognition [4] (CVPR 2015) and open-world continual learning [10] (arXiv 2023) add rejection. Agent memories with consolidation [16, 17] (UIST 2023, arXiv 2023) and description-based classification [14] (ICLR 2023) motivate our memory notes.

## 3. Preliminaries

Setting. After session $t$ the released inventory is $C_t = C_{t-1} \cup \Delta C_t$; each new class comes with $K = 5$ labelled supports $S_t$. The model is frozen, $\theta_t = \theta_0$, and only the memory evolves:

$$M_t = \mathcal{C}_B\big(M_{t-1} \cup E(S_t)\big), \qquad \text{bytes}(M_t) \le B = 131{,}072, \tag{1}$$

where $E$ writes a note with the frozen model from a class's own supports and $\mathcal{C}_B$ evicts stored references until the budget holds. A query is answered $\hat{y} \in C_t \cup \{\text{UNKNOWN}\}$ by the model. Future and held-out classes are never named, shown or stored.

Decision rule. Every arm ends with one constrained call listing the released classes and UNKNOWN as letters; the label is the letter with the highest next-token probability. Known-ness is $1 - p(\text{UNKNOWN})$. Open-world balanced accuracy (BA, primary) counts UNKNOWN on a known image as an error; forced-choice BA removes UNKNOWN; calibrated BA answers UNKNOWN only below the known-ness that keeps 95% of known development images.

Data and cohorts. Locked CLOVE-Bench manifests (group-aware, de-duplicated by SHA-256 and perceptual hash) fix supports and five class orders. Development queries come from the locked development split; confirmatory (fresh) queries come from source groups used by no earlier experiment, at most one per group. Table 1 lists every candidate data set and its status.

## 4. CLOVE: Memory and Orchestration

Uniform memory (B2). A class record holds a note ($\leq$48 tokens), $224^2$ JPEG thumbnails (quality 70) of retained supports and an fp16 index vector each; eviction removes thumbnails from the class with most thumbnails. A $224^2$ textured field image costs about 17 KB, so under 128 KiB the memory keeps about one image per class or fewer once ten or more classes are released (Fig. 4a).

Dense memory (D). Same budget and eviction rule, but $128^2$ thumbnails (quality 55, $\approx$5 KB) and an int8 code of a fixed random projection of the frozen vision feature to 128 dimensions, regenerated from a stored seed (130 bytes per thumbnail instead of 4 KB). The index only orders evidence; the model makes every decision.

Orchestration. B0 zero-shot; B1 two-shot in-context learning (the first two of the five supports per class, all released classes in one prompt; not byte-budgeted); B2 one prompt with the memory; B3 fixed SAGE-style orchestration (text shortlist, then inspection of three classes); B4 autonomous orchestration (the model chooses which classes to inspect); B12 retrieve-then-verify (the index shortlists three classes, the model chooses among them or UNKNOWN); C1 exhaustive independent verification of every class followed by a final call. CLOVE-D12 is retrieve-then-verify on the dense memory; it was selected among D2, D3, D4 and D12 by a pre-stated rule on development data only (DeepWeeds and Mango).

## 5. Experiments

Setup. Qwen2.5-VL-3B-Instruct [3] (arXiv 2025) (revision 66285546, fp16) on two 12 GB GPUs; at most 12 thumbnails per prompt. Statistics: class-stratified paired bootstrap over source groups (orders crossed when several are pooled), exact McNemar, Holm within each pre-registered family. A contrast is confirmed with $\Delta \geq 3$ points, 95% CI above zero and $p_{\text{Holm}} < 0.05$.

Frozen-feature reference. Table 2 reports continual readouts on the same Qwen vision features and manifests (closed-set accuracy on the locked test split, final session). They decide labels from features and are therefore not CLOVE arms, but they show what the frozen representation supports.

### 5.1. Generative collapse

At the last DeepWeeds development session the true class is in the index top three for 80% of images, but the model's forced choice is correct for 43% with the memory, 33% with two-shot in-context learning and 40% when only the three retrieved classes are offered (Fig. 1d, Fig. 2). The most-predicted label receives 33% of known images where 17% is expected, and reversing the option order preserves only 50% of answers. A blank query image yields UNKNOWN for every known image and permuting class names against the evidence lowers accuracy, so the evidence is used.

**Observation**

Coverage is not the problem: the answer is retrieved, but the frozen model does not pick it.

### 5.2. Agentic orchestration under pre-registered confirmation

On never-examined groups of DeepWeeds and Dangerous Insects, registered before any fresh image was evaluated, exhaustive verification (C1) is not confirmed, retrieve-then-verify (B12) is not confirmed and two-shot in-context learning (B1) is not confirmed relative to the uniform memory (Fig. 3, upper block).

### 5.3. Memory optimisation

On development data the dense memory lifts DeepWeeds forced choice when combined with retrieve-then-verify, and the pre-stated rule selected D12. On never-examined groups (Table 3) the pre-registered rule is met on 0 of 14 data sets (none). Calibrated rejection of held-out classes is reported alongside, because the dense arm answers UNKNOWN more often in open-world mode.

Table 1. Data-set coverage: all twelve official AgEval subsets, further identification sets of the AgEval loader, and external benchmarks. “evaluated”: part of the pre-registered confirmation.

| Data set | Classes | Source | Status | Note |
|---|---|---|---|---|
| AgEval | | | | |
| Soybean Seeds | 5 | Kaggle (AgEval loader) | evaluated | |
| Mango Leaf Disease | 8 | Kaggle MangoLeafBD (AgEval loader) | development only | almost no source groups outside earlier partitions |
| DeepWeeds | 8 | Kaggle DeepWeeds (AgEval loader) | evaluated | Negative class excluded |
| Bean Leaf Lesions | 3 | Kaggle (AgEval loader) | evaluated | incremental only (no held-out class) |
| Dangerous Insects | 15 | Kaggle (AgEval loader) | evaluated | |
| Soybean Diseases (PNAS) | 7 | Zenodo 12747481 (AgEval loader) | evaluated | folders 1/7 excluded (label-map conflict) |
| Durum Wheat | 3 | Kaggle (AgEval loader) | excluded | each class is frames of one video: 5 independent supports impossible |
| Yellow Rust 19 | 6 | AgEval loader | evaluated | severity grades as classes (ordinal caveat); RAW photos, augmented copies avoided |
| FUSARIUM 22 | 5 | AgEval loader | evaluated | wilt grades as classes (ordinal caveat) |
| IDC (iron deficiency chlorosis) | 4 | AgEval loader | development only | severity ratings; ratings 3-4 have no never-examined groups; rating 5 insufficient |
| PlantDoc (segmentation) | – | AgEval loader | excluded | segmentation/quantification task: no class labels |
| InsectCount | – | AgEval loader | excluded | counting task: no class labels |
| AgEval loader | | | | |
| Rice Leaf Disease Severity (SBRD) | 9 | Kaggle (AgEval loader) | evaluated | |
| IP102 (25 classes) | 25 | Kaggle IP102 (AgEval loader) | evaluated | 25 hash-selected classes |
| external | | | | |
| iNatAg-25 | 25 | HF Project-AgML/iNatAg-mini | evaluated | |
| iNatAg-100 | 100 | HF Project-AgML/iNatAg-mini | evaluated | pre-registered >=25-class extension |
| PlantDoc (classification) | 27 | GitHub pratikkayal/PlantDoc-Dataset (CC BY 4.0) | evaluated | web images; spider-mite class excluded (2 images) |
| PlantVillage | 38 | HF mohanty/PlantVillage (CC BY-SA 3.0) | evaluated | grouped by leaf-map; 26% ungrouped |
| IP102 (96 classes) | 96 | Kaggle IP102 | evaluated | 6 classes with <60 images excluded; overlap with IP102-25 cohorts excluded |
| PlantWild | 89 | HF uqtwei2/PlantWild (CC BY-NC-ND 4.0) | development only | 40/class sample used every source group; PlantWild-v2 not used (no split metadata) |
| iNatAg-500 | 500 | HF Project-AgML/iNatAg-mini | not run | 46 GB stream exceeds the run budget |

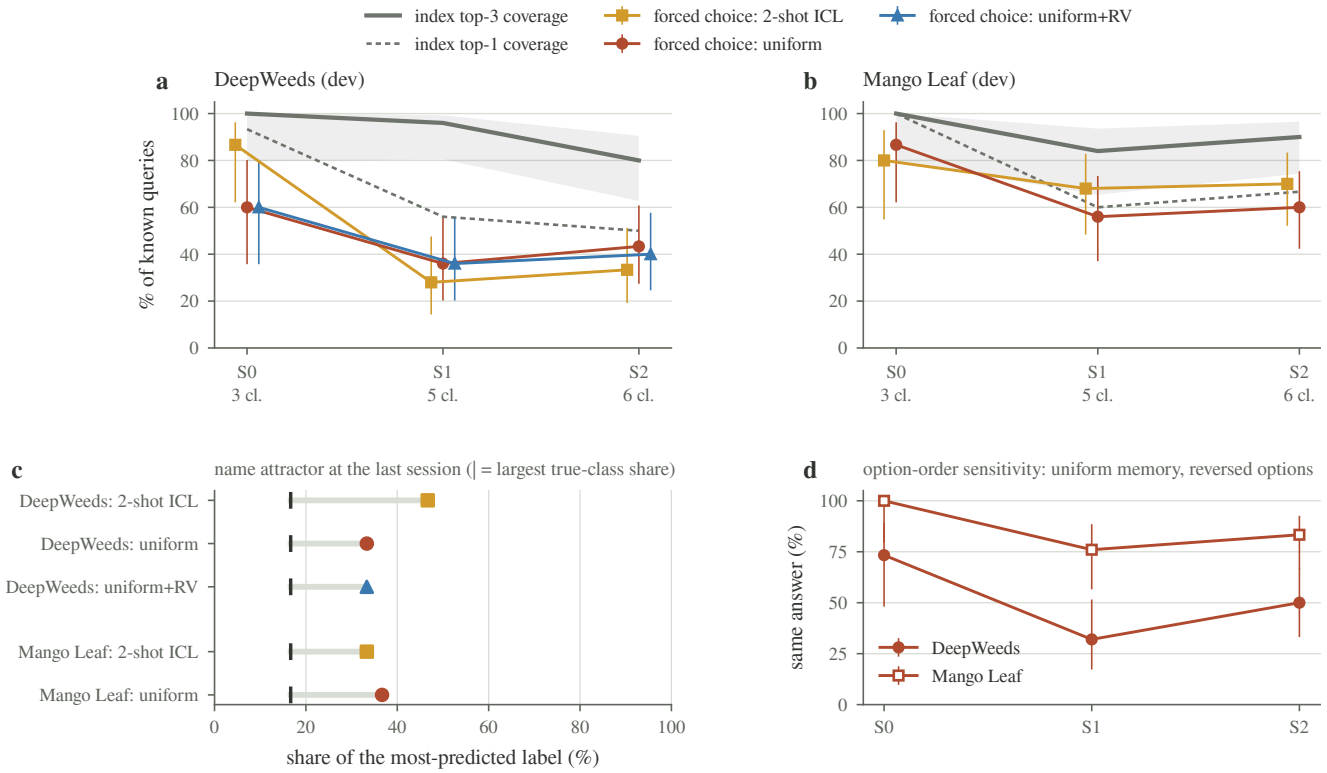



Figure 2. Does the model fail because evidence is missing, or because it does not use it? Development cohort, class order 42. (a,b) Grey: retrieval quantities (share of known queries whose class is in the memory index’s top-1 / top-3; band: Wilson 95% interval for top-3). Colour: the model’s forced-choice accuracy on the same queries (Wilson 95%). (c) Share of known queries assigned to the most-predicted label at the last session, against the largest true-class share. (d) Agreement of the uniform-memory answer with the answer under reversed option order.

### 5.4. Data set by data set

Table 4 and Fig. 5 place all data sets on one axis: how many classes compete for the 128 KiB budget. Where few classes are released the uniform memory already holds two to three images per class and the dense memory adds little or hurts; where twenty or more classes are released the uniform memory falls below one image per class.

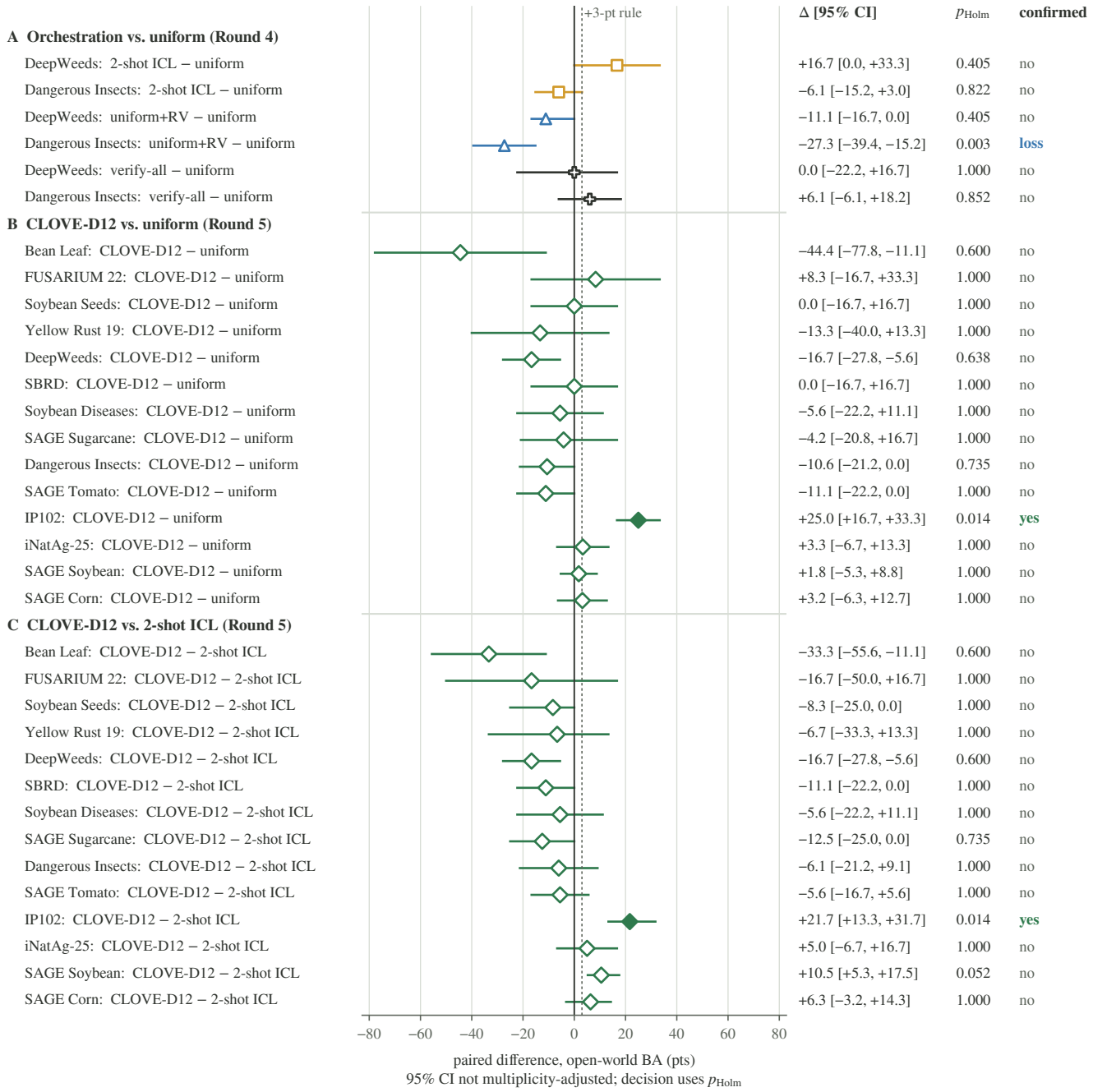



Figure 3. Every confirmatory contrast, by inferential family. Paired difference in final-session open-world balanced accuracy on never-examined image groups with 95% bootstrap CIs that are not multiplicity-adjusted; the decision uses Holm-adjusted $p$ within each pre-registered family (A: the six Round-4 orchestration tests; B and C together: one family of all Round-5 order-42 tests). "yes": $\geq$3 points, CI above zero and $p_{\text{Holm}} < 0.05$; "loss": CI below zero and $p_{\text{Holm}} < 0.05$. An unadjusted CI excluding zero does not by itself imply significance after correction: several Round-5 contrasts on few-class data sets have CIs below zero without surviving Holm correction; exact McNemar tests on their 9–30 paired units are also non-significant.

**Key finding**

Memory optimisation pays off only where the budget is the bottleneck (PlantDoc (+27 points vs B12); PlantVillage (+27 points vs B12); iNatAg-100 (+9 points vs B12)); on few-class data sets the uniform memory already fits and the dense memory adds noise.

## 6. Limitations and Conclusion

Limitations. Query sets per data set are small and the primary test uses one class order; further orders

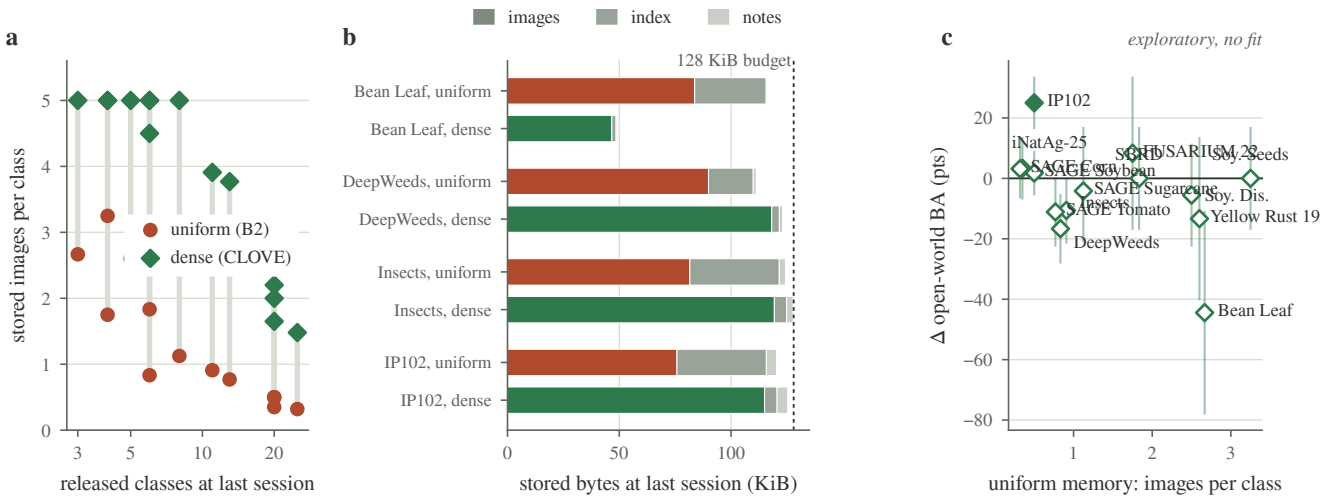



Figure 4. Fixed-budget memory analysis (128 KiB; no budget sweep is reported). (a) Measured stored images per class at the last session (order 42) for each data set, against the number of released classes. (b) Measured byte composition; on Bean Leaf the dense memory holds every support well below the budget, so its loss there is not a budget effect. (c) Exploratory: CLOVE-D12 effect against how starved the uniform memory is (no fit; too few independent data sets for a trend claim).

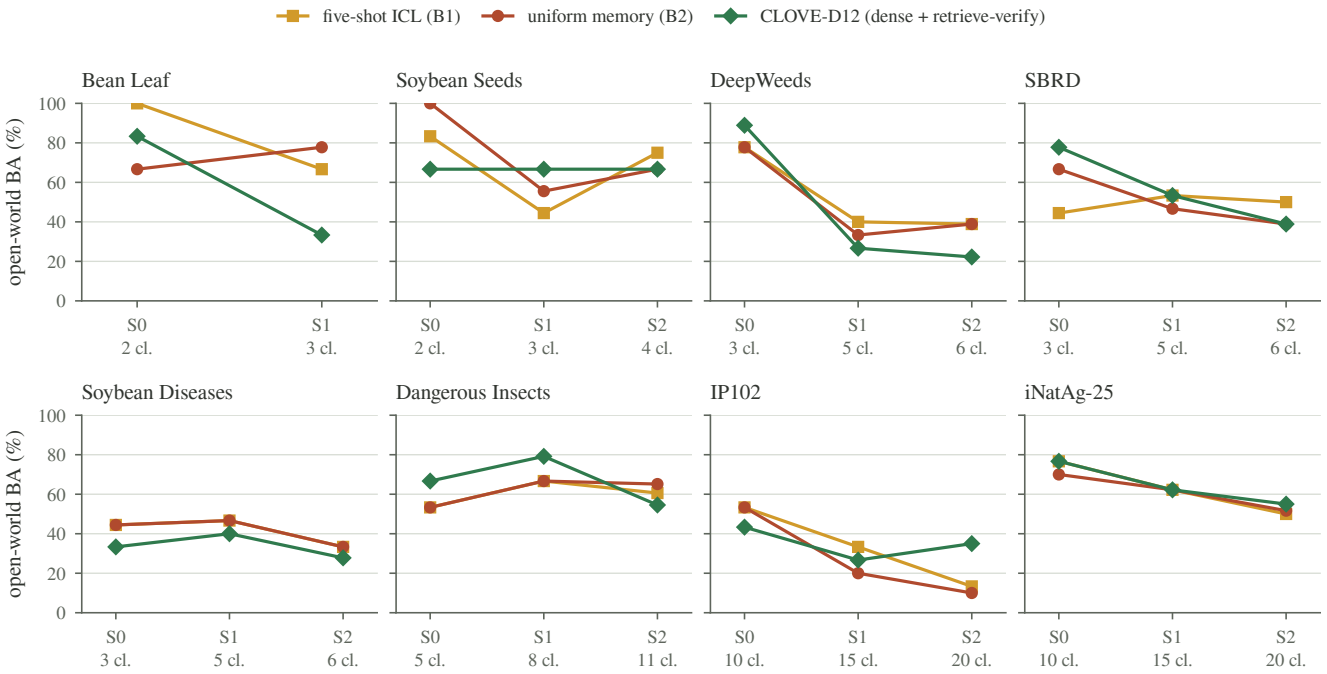



Figure 5. Continual learning curves. Open-world balanced accuracy at every session on never-examined image groups, one class order (42); the tick labels give the number of released classes. Bean Leaf has two sessions. Colours and markers are fixed per method in every figure.

are pooled where complete. One 3B model in fp16 was used. Mango Leaf serves for development only. Pretraining exposure to these public data sets cannot be excluded. The many-class hypothesis was formed after the IP102 result; its extension to iNatAg-100 is registered separately. Monolithic prompts are limited to 25 classes by the single-token option alphabet.

Not completed. Memory-budget sweeps (32 KB–1 MB), shot-count sweeps ($K \in \{1, 2, 4, 8, 16\}$),

Table 2. Frozen-feature continual readouts on the same Qwen vision features (closed-set accuracy, %, final session, locked test split; best in bold). NCM: nearest class mean [21] (arXiv 2019); SLDA [8] (CVPRW 2020); FeCAM [7], RanPAC [13], TEEN [20] (all NeurIPS 2023); Linear: full-memory linear probe.

| Data set | NCM | 1-NN | SLDA | SLDA-r8 | FeCAM | RanPAC | TEEN | Linear |
|---|---|---|---|---|---|---|---|---|
| Bean Leaf | 76 | 69 | 82 | 76 | 73 | 77 | 68 | 75 |
| DeepWeeds | 72 | 68 | 74 | 72 | 74 | 76 | 68 | 74 |
| Dangerous Insects | 70 | 65 | 72 | 71 | 64 | 73 | 66 | 72 |
| Mango Leaf | 82 | 84 | 93 | 92 | 89 | 90 | 77 | 82 |
| SBRD | 68 | 72 | 74 | 73 | 72 | 75 | 65 | 71 |
| Soybean Seeds | 66 | 52 | 66 | 70 | 59 | 68 | 61 | 66 |
| iNatAg-25 | 78 | 77 | 83 | 83 | 77 | 83 | 76 | 82 |

domain-shift tracks, the remaining AgEval severity subsets and external data sets (PlantDoc, PlantVillage) were not completed within the compute budget; no result is reported for them.

Conclusion. A frozen 3B VLM with a 128 KiB external memory learns new agricultural categories from five images while few categories are known, but its own decision collapses as categories accumulate even when retrieval covers the answer. Agentic inspection and verification do not repair this. Optimising what the memory stores helps only where the byte budget binds, and frozen-feature readouts remain stronger in closed-set accuracy. Closing the gap between what the memory contains and what the model decides is the open problem.

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Table 3. CLOVE-D12 against the uniform memory (B2) and two-shot ICL (B1) on never-examined image groups (final session, mean over completed class orders). BA open / forced / calibrated, old / new classes, held-out rejection (raw and calibrated), known-ness AUROC, model calls per query.

| Arm | BA | $BA_f$ | $BA_{cal}$ | Old | New | Rej. | $Rej._{cal}$ | AUROC | Calls |
|---|---|---|---|---|---|---|---|---|---|
| DeepWeeds | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 39 | 50 | 41 | 38 | 44 | 39 | 11 | 0.60 | 1.0 |
| B2 uniform memory | 41 | 44 | 38 | 40 | 44 | 39 | 28 | 0.64 | 1.0 |
| D12 dense + retrieve-then-verify | 26 | 48 | 36 | 27 | 22 | 72 | 56 | 0.71 | 1.0 |
| Dangerous Insects | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 61 | 67 | 61 | 67 | 44 | 67 | 58 | 0.83 | 1.0 |
| B2 uniform memory | 65 | 68 | 65 | 67 | 61 | 75 | 75 | 0.87 | 1.0 |
| D12 dense + retrieve-then-verify | 55 | 58 | 58 | 58 | 44 | 92 | 92 | 0.83 | 1.0 |
| SBRD | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 58 | 64 | 59 | 50 | 100 | 17 | 0 | 0.58 | 1.0 |
| B2 uniform memory | 53 | 53 | 50 | 53 | 50 | 0 | 0 | 0.52 | 1.0 |
| D12 dense + retrieve-then-verify | 44 | 58 | 52 | 40 | 67 | 67 | 39 | 0.77 | 1.0 |
| Soybean Seeds | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 71 | 79 | 80 | 67 | 83 | 50 | 0 | 0.78 | 1.0 |
| B2 uniform memory | 71 | 71 | 77 | 61 | 100 | 0 | 0 | 0.62 | 1.0 |
| D12 dense + retrieve-then-verify | 58 | 67 | 60 | 50 | 83 | 33 | 33 | 0.78 | 1.0 |
| Bean Leaf | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 56 | 56 | 56 | 58 | 50 | — | — | — | 1.0 |
| B2 uniform memory | 72 | 72 | 72 | 83 | 50 | — | — | — | 1.0 |
| D12 dense + retrieve-then-verify | 50 | 50 | 50 | 75 | 0 | — | — | — | 1.0 |
| iNatAg-25 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 53 | 56 | 56 | 54 | 49 | 36 | 4 | 0.67 | 1.0 |
| B2 uniform memory | 50 | 52 | 53 | 56 | 31 | 44 | 4 | 0.70 | 1.0 |
| D12 dense + retrieve-then-verify | 56 | 62 | 64 | 56 | 53 | 58 | 7 | 0.67 | 1.0 |
| IP102 (25 cl.) | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 25 | 26 | 26 | 24 | 29 | 49 | 18 | 0.66 | 1.0 |
| B2 uniform memory | 14 | 16 | 15 | 16 | 11 | 67 | 16 | 0.63 | 1.0 |
| D12 dense + retrieve-then-verify | 31 | 48 | 43 | 27 | 44 | 69 | 36 | 0.66 | 1.0 |
| Soybean Diseases (7 cl.) | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 36 | 39 | 38 | 33 | 50 | 0 | 0 | 0.74 | 1.0 |
| B2 uniform memory | 33 | 33 | 33 | 30 | 50 | 0 | 0 | 0.89 | 1.0 |
| D12 dense + retrieve-then-verify | 39 | 50 | 43 | 30 | 83 | 17 | 0 | 0.54 | 1.0 |
| iNatAg-100 | | | | | | | | | |
| D12 dense + retrieve-then-verify | 15 | 16 | 16 | 15 | 14 | 65 | 8 | 0.62 | 1.0 |
| B12 uniform + retrieve-then-verify | 6 | 6 | 6 | 6 | 0 | 73 | 4 | 0.46 | 1.0 |
| PlantDoc | | | | | | | | | |
| D12 dense + retrieve-then-verify | 47 | 53 | 52 | 44 | 58 | 20 | 7 | 0.56 | 1.0 |
| B12 uniform + retrieve-then-verify | 20 | 20 | 20 | 24 | 0 | 27 | 0 | 0.48 | 1.0 |
| PlantVillage | | | | | | | | | |
| D12 dense + retrieve-then-verify | 38 | 46 | 43 | 36 | 47 | 33 | 0 | 0.45 | 1.0 |
| B12 uniform + retrieve-then-verify | 11 | 16 | 16 | 12 | 7 | 38 | 0 | 0.39 | 1.0 |
| IP102 (96 cl.) | | | | | | | | | |
| D12 dense + retrieve-then-verify | 7 | 10 | 10 | 8 | 5 | 61 | 11 | 0.47 | 1.0 |
| B12 uniform + retrieve-then-verify | 3 | 4 | 4 | 3 | 0 | 72 | 17 | 0.46 | 1.0 |
| Yellow Rust 19 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 40 | 40 | 33 | 50 | 0 | 0 | 0 | 0.84 | 1.0 |
| B2 uniform memory | 47 | 47 | 40 | 58 | 0 | 0 | 0 | 0.82 | 1.0 |
| D12 dense + retrieve-then-verify | 33 | 33 | 27 | 33 | 33 | 100 | 33 | 1.00 | 1.0 |
| FUSARIUM 22 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 50 | 50 | 42 | 44 | 67 | 0 | 0 | 0.58 | 1.0 |
| B2 uniform memory | 25 | 25 | 25 | 11 | 67 | 0 | 0 | 0.36 | 1.0 |
| D12 dense + retrieve-then-verify | 33 | 42 | 33 | 44 | 0 | 67 | 0 | 0.44 | 1.0 |
| SAGE Corn | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 19 | 19 | 19 | 13 | 56 | 0 | 0 | 0.64 | 1.0 |

Table 4. Storyline across data sets (never-examined groups, final session). $|C_T|$ released classes; Refs: stored images per class in the uniform (U) and dense (D) memory; BA: open-world balanced accuracy (%), mean over completed orders; $\Delta$: paired CLOVE-D12 effect on order 42 with unadjusted 95% CI; verdict uses $p_{\text{Holm}}$ and the pre-registered rule. Total: classes in the locked subset; Held: never-released classes; —: not applicable (arm not run).

| Data set | Total | Held | $\lvert C_T\rvert$ | Refs U | Refs D | B1 | B2 | B12 | D12 | $D12_f$ | $\Delta$ (pts) | $p_{\text{Holm}}$ | Verdict |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bean Leaf | 3 | 0 | 3 | 2.7 | 5.0 | 56 | 72 | — | 50 | 50 | −44 [−78, −11] (vs B2) | 0.600 | lower (n.s. after Holm) |
| FUSARIUM 22 | 5 | 1 | 4 | 1.8 | 5.0 | 50 | 25 | — | 33 | 42 | +8 [−17, +33] (vs B2) | 1.000 | no detectable difference |
| Soybean Seeds | 5 | 1 | 4 | 3.2 | 5.0 | 71 | 71 | — | 58 | 67 | +0 [−17, +17] (vs B2) | 1.000 | no detectable difference |
| Yellow Rust 19 | 6 | 1 | 5 | 2.6 | 5.0 | 40 | 47 | — | 33 | 33 | −13 [−40, +13] (vs B2) | 1.000 | no detectable difference |
| DeepWeeds | 8 | 2 | 6 | 0.8 | 4.5 | 39 | 41 | — | 26 | 48 | −17 [−28, −6] (vs B2) | 0.638 | lower (n.s. after Holm) |
| SBRD | 9 | 3 | 6 | 1.8 | 5.0 | 58 | 53 | — | 44 | 58 | +0 [−17, +17] (vs B2) | 1.000 | no detectable difference |
| Soybean Diseases (7 cl.) | 7 | 1 | 6 | 2.5 | 5.0 | 36 | 33 | — | 39 | 50 | −6 [−22, +11] (vs B2) | 1.000 | no detectable difference |
| SAGE Sugarcane | 12 | 2 | 8 | 1.1 | 5.0 | 50 | 42 | 38 | 38 | 54 | −4 [−21, +17] (vs B2) | 1.000 | no detectable difference |
| Dangerous Insects | 15 | 4 | 11 | 0.9 | 3.9 | 61 | 65 | — | 55 | 58 | −11 [−21, −0] (vs B2) | 0.735 | lower (n.s. after Holm) |
| SAGE Tomato | 17 | 3 | 13 | 0.8 | 3.8 | 14 | 19 | 14 | 8 | 19 | −11 [−22, +0] (vs B2) | 1.000 | no detectable difference |
| IP102 (25 cl.) | 25 | 5 | 20 | 0.5 | 2.2 | 25 | 14 | — | 31 | 48 | +25 [+17, +33] (vs B2) | 0.014 | accuracy gain; rejection clause not met |
| iNatAg-25 | 25 | 5 | 20 | 0.3 | 1.6 | 53 | 50 | — | 56 | 62 | +3 [−7, +13] (vs B2) | 1.000 | no detectable difference |
| SAGE Soybean | 28 | 6 | 20 | 0.5 | 2.0 | 4 | 12 | 12 | 14 | 21 | +2 [−5, +9] (vs B2) | 1.000 | no detectable difference |
| PlantDoc | 27 | 5 | 22 | 0.4 | 1.9 | — | — | 20 | 47 | 53 | +27 [+18, +36] (vs B12) | 0.003 | confirmed gain |
| SAGE Corn | 32 | 6 | 25 | 0.3 | 1.5 | 19 | 22 | 16 | 25 | 33 | +3 [−6, +13] (vs B2) | 1.000 | no detectable difference |
| PlantVillage | 38 | 8 | 30 | 0.3 | 1.6 | — | — | 11 | 38 | 46 | +27 [+18, +36] (vs B12) | 0.003 | confirmed gain |
| IP102 (96 cl.) | 96 | 18 | 78 | 0.1 | 0.5 | — | — | 3 | 7 | 10 | +5 [+2, +8] (vs B12) | 0.006 | accuracy gain; rejection clause not met |
| iNatAg-100 | 100 | 20 | 80 | 0.1 | 0.4 | — | — | 6 | 15 | 16 | +9 [+6, +12] (vs B12) | 0.003 | confirmed gain |

## A. Dataset-by-Dataset Account

Bean Leaf (3 released classes at the last session; the uniform memory keeps 2.7 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 56% with two-shot in-context learning, 72% with the uniform memory and 50% with D12 (forced choice 72% vs 50%). D12−B2 on order 42: −44 [−78, −11] points, $p_{\text{Holm}} = 0.600$: lower (n.s. after Holm).

FUSARIUM 22 (4 released classes at the last session; the uniform memory keeps 1.8 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 50% with two-shot in-context learning, 25% with the uniform memory and 33% with D12 (forced choice 25% vs 42%). D12−B2 on order 42: +8 [−17, +33] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 0% (D12).

Soybean Seeds (4 released classes at the last session; the uniform memory keeps 3.2 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 71% with two-shot in-context learning, 71% with the uniform memory and 58% with D12 (forced choice 71% vs 67%). D12−B2 on order 42: +0 [−17, +17] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 33% (D12).

Yellow Rust 19 (5 released classes at the last session; the uniform memory keeps 2.6 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 40% with two-shot in-context learning, 47% with the uniform memory and 33% with D12 (forced choice 47% vs 33%). D12−B2 on order 42: −13 [−40, +13] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 33% (D12).

DeepWeeds (6 released classes at the last session; the uniform memory keeps 0.8 and the dense memory 4.5 thumbnails per class). Open-world balanced accuracy is 39% with two-shot in-context learning, 41% with the uniform memory and 26% with D12 (forced choice 44% vs 48%). D12−B2 on order 42: −17 [−28, −6] points, $p_{\text{Holm}} = 0.638$: lower (n.s. after Holm). Calibrated held-out rejection: 28% (B2) vs 56% (D12).

SBRD (6 released classes at the last session; the uniform memory keeps 1.8 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 58% with two-shot in-context learning, 53% with the uniform memory and 44% with D12 (forced choice 53% vs 58%). D12−B2 on order 42: +0 [−17, +17] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 39% (D12).

Soybean Diseases (7 cl.) (6 released classes at the last session; the uniform memory keeps 2.5 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 36% with two-shot in-context learning, 33% with the uniform memory and 39% with D12 (forced choice 33% vs 50%). D12−B2 on order 42: −6 [−22, +11] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 0% (D12).

SAGE Sugarcane (8 released classes at the last session; the uniform memory keeps 1.1 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 50% with two-shot in-context learning, 42% with the uniform memory and 38% with D12 (forced choice 42% vs 54%). D12−B2 on order 42: −4 [−21, +17] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 0% (D12).

Dangerous Insects (11 released classes at the last session; the uniform memory keeps 0.9 and the dense memory 3.9 thumbnails per class). Open-world balanced accuracy is 61% with two-shot in-context learning, 65% with the uniform memory and 55% with D12 (forced choice 68% vs 58%). D12−B2 on order 42: −11 [−21, −0] points, $p_{\text{Holm}} = 0.735$: lower (n.s. after Holm). Calibrated held-out rejection: 75% (B2) vs 92% (D12).

SAGE Tomato (13 released classes at the last session; the uniform memory keeps 0.8 and the dense memory 3.8 thumbnails per class). Open-world balanced accuracy is 14% with two-shot in-context learning, 19% with the uniform memory and 8% with D12 (forced choice 19% vs 19%). D12−B2 on order 42: −11 [−22, +0] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 33% (B2) vs 0% (D12).

IP102 (25 cl.) (20 released classes at the last session; the uniform memory keeps 0.5 and the dense memory 2.2 thumbnails per class). Open-world balanced accuracy is 25% with two-shot in-context learning, 14% with the uniform memory and 31% with D12 (forced choice 16% vs 48%). D12−B2 on order 42: +25 [+17, +33] points, $p_{\text{Holm}} = 0.014$: accuracy gain; rejection clause not met. Calibrated held-out rejection: 16% (B2) vs 36% (D12).

iNatAg-25 (20 released classes at the last session; the uniform memory keeps 0.3 and the dense memory 1.6 thumbnails per class). Open-world balanced accuracy is 53% with two-shot in-context learning, 50% with the uniform memory and 56% with D12 (forced choice 52% vs 62%). D12−B2 on order 42: +3 [−7, +13] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 4% (B2) vs 7% (D12).

SAGE Soybean (20 released classes at the last session; the uniform memory keeps 0.5 and the dense memory 2.0 thumbnails per class). Open-world balanced accuracy is 4% with two-shot in-context learn-

ing, 12% with the uniform memory and 14% with D12 (forced choice 12% vs 21%). D12−B2 on order 42: +2 [−5, +9] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 17% (B2) vs 33% (D12).

PlantDoc (22 released classes at the last session; the uniform memory keeps 0.4 and the dense memory 1.9 thumbnails per class). Open-world balanced accuracy is 20% with retrieve-then-verify on the uniform memory and 47% on the dense memory. D12−B12 on order 42: +27 [+18, +36] points, $p_{\text{Holm}} = 0.003$: confirmed gain.

SAGE Corn (25 released classes at the last session; the uniform memory keeps 0.3 and the dense memory 1.5 thumbnails per class). Open-world balanced accuracy is 19% with two-shot in-context learning, 22% with the uniform memory and 25% with D12 (forced choice 22% vs 33%). D12−B2 on order 42: +3 [−6, +13] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 17% (D12).

PlantVillage (30 released classes at the last session; the uniform memory keeps 0.3 and the dense memory 1.6 thumbnails per class). Open-world balanced accuracy is 11% with retrieve-then-verify on the uniform memory and 38% on the dense memory. D12−B12 on order 42: +27 [+18, +36] points, $p_{\text{Holm}} = 0.003$: confirmed gain.

IP102 (96 cl.) (78 released classes at the last session; the uniform memory keeps 0.1 and the dense memory 0.5 thumbnails per class). Open-world balanced accuracy is 3% with retrieve-then-verify on the uniform memory and 7% on the dense memory. D12−B12 on order 42: +5 [+2, +8] points, $p_{\text{Holm}} = 0.006$: accuracy gain; rejection clause not met.

iNatAg-100 (80 released classes at the last session; the uniform memory keeps 0.1 and the dense memory 0.4 thumbnails per class). Open-world balanced accuracy is 6% with retrieve-then-verify on the uniform memory and 15% on the dense memory. D12−B12 on order 42: +9 [+6, +12] points, $p_{\text{Holm}} = 0.003$: confirmed gain.

## B. Development Results and Gap Analysis

## C. Replication over Class Orders

## D. Round-4 Confirmation Details

The confirmatory run was registered before any fresh image was evaluated (rounds/R4/preregistration.md): DeepWeeds and Dangerous Insects, class order 42, three known and three held-out queries per class and session from source groups never examined before, arms B0, B1, B2, B12 and C1, primary endpoint final-session open-world balanced accuracy, and the family {C1−B2, B12−B2, B1−B2} on both data sets with Holm correction. A contrast is confirmed only with $\Delta \geq 3$ points, a 95% interval above zero, $p_{\text{Holm}} < 0.05$ on both data sets, and no more than 10 points lower held-out rejection. Under this rule independent verification (C1) is not confirmed, retrieve-then-verify (B12) is not confirmed and five-shot in-context learning (B1) is not confirmed relative to the uniform memory (B2). Tables 9 and 10 and Figure 6 give all values.

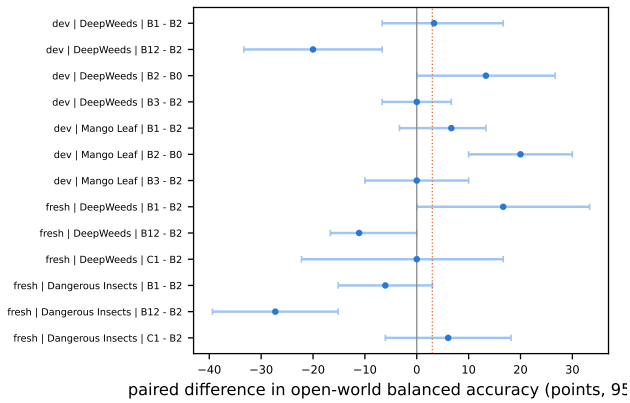



Figure 6. Paired differences in open-world balanced accuracy with 95% intervals (development and confirmatory). Dotted line: the pre-registered +3-point threshold.

## E. Round-5 Details

## F. Step 3: Memory Optimisation (Round 5)

The development diagnostics show that the 128 KiB uniform memory keeps 0–1 reference image per DeepWeeds class, because a $224^2$ JPEG of a textured weed image is about 17 KB and its fp16 index another 4 KB. The dense memory keeps the same budget and eviction rule but stores $128^2$ thumbnails (quality 55, about 5 KB) and replaces the fp16 index by an int8 code of a fixed random projection to 128 dimensions that is regenerated from a stored seed (130 bytes per thumbnail). The index orders and shortlists evidence only; the model chooses every label. We combine it with the three orchestration designs: D2 (monolithic), D3 (fixed SAGE-style), D4 (autonomous) and D12 (retrieve-then-verify), with B2 (uniform memory) and B1 (five-shot ICL on raw supports) in the same runs. A secondary, pre-declared calibrated read-out takes the model's forced choice and answers UNKNOWN only when its known-ness falls below the threshold that keeps 95% of known development images of the same arm and data set.

Table 5. Development cohort, class order 42, final session (columns as in Table 3; index-only row is not a model decision).

| Arm | BA | $BA_f$ | Old | New | Rej. | FR | AUROC | Calls |
|---|---|---|---|---|---|---|---|---|
| Mango Leaf, final session 2, 30 known / 9 held-out queries | | | | | | | | |
| B0 zero-shot | 40 | 40 | 28 | 100 | 78 | 10 | 0.80 | 1.0 |
| B1 five-shot ICL | 67 | 70 | 60 | 100 | 11 | 3 | 0.79 | 1.0 |
| B2 uniform memory | 60 | 60 | 52 | 100 | 22 | 0 | 0.85 | 1.0 |
| B3 fixed orchestration | 60 | 63 | 52 | 100 | 11 | 7 | 0.63 | 2.0 |
| control: blank query | 0 | 17 | 0 | 0 | 100 | 100 | 0.00 | 1.0 |
| control: permuted names | 40 | 43 | 44 | 20 | 0 | 3 | 0.75 | 1.0 |
| control: reversed options | 60 | 63 | 52 | 100 | 22 | 3 | 0.83 | 1.0 |
| DeepWeeds, final session 2, 30 known / 10 held-out queries | | | | | | | | |
| B12 retrieve-then-verify | 10 | 40 | 12 | 0 | 100 | 83 | 0.65 | 1.0 |
| B13 one-vs-one yes/no | 17 | 37 | 8 | 60 | 90 | 70 | 0.65 | 3.0 |
| B0 zero-shot | 17 | 27 | 20 | 0 | 50 | 17 | 0.85 | 1.0 |
| B1 five-shot ICL | 33 | 33 | 40 | 0 | 10 | 10 | 0.60 | 1.0 |
| B2 uniform memory | 30 | 43 | 28 | 40 | 60 | 37 | 0.73 | 1.0 |
| B3 fixed orchestration | 30 | 37 | 28 | 40 | 60 | 53 | 0.57 | 2.0 |
| control: blank query | 0 | 17 | 0 | 0 | 100 | 100 | 0.00 | 1.0 |
| control: permuted names | 13 | 23 | 16 | 0 | 80 | 50 | 0.73 | 1.0 |
| control: reversed options | 30 | 40 | 28 | 40 | 70 | 50 | 0.74 | 1.0 |
| index-only top-1 (reference, not Qwen) | 50 | – | – | – | – | – | – | 0 |

Table 6. Coverage versus decision (development, %).

| Data set | $t$ | $\lvert C_t \rvert$ | Cov@1 | Cov@3 | B1 | B2 | B12 | B2\|cov | Attractor | Expected | Rev. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Mango Leaf | 0 | 3 | 100 | 100 | 80 | 87 | – | 87 | 47 | 33 | 100 |
| Mango Leaf | 1 | 5 | 60 | 84 | 68 | 56 | – | 53 | 40 | 20 | 76 |
| Mango Leaf | 2 | 6 | 67 | 90 | 70 | 60 | – | 61 | 37 | 17 | 83 |
| DeepWeeds | 0 | 3 | 93 | 100 | 87 | 60 | 60 | 60 | 60 | 33 | 73 |
| DeepWeeds | 1 | 5 | 56 | 96 | 28 | 36 | 36 | 35 | 64 | 20 | 32 |
| DeepWeeds | 2 | 6 | 50 | 80 | 33 | 43 | 40 | 48 | 33 | 17 | 50 |

Confirmation. The selected arm D12 was compared with B2 and B1 on never-examined image groups of DeepWeeds, Dangerous Insects, SBRD, Soybean Seeds, iNatAg-25 and Bean Leaf, plus the two AgEval data sets added afterwards, IP102 (25 classes) and Soybean Diseases (7 classes) (class order 42; calibration thresholds from a separate known-class development draw). Under the pre-registered rule (open-world balanced accuracy, $\geq 3$ points, interval above zero, $p_{\text{Holm}} < 0.05$, on at least two data sets) the improvement over B2 is met on 0 of 14 data sets (none). Table 14 gives all values, Table 15 all contrasts.

Data sets with at least 25 classes. The only data set meeting the rule above is the largest one, IP102. After seeing this, and before running it, we pre-registered an extension (rounds/R5/hypotheses_25plus.md): dense memory should help when the 128 KiB budget is spread over many classes. It covers every locked manifest with at least 25 classes (IP102, iNatAg-25, iNatAg-100) and adds B12, the same retrieve-then-verify orchestration on the uniform memory, so that D12−B12 isolates the memory optimisation at equal orchestration. B2 and B1 cannot be run on iNatAg-100: 40–80 released classes exceed the single-token option alphabet of the decision rule. iNatAg-100 query images that appear in any iNatAg-25 cohort are excluded. Only iNatAg-100 and class orders 43/44 are confirmatory here; IP102 and iNatAg-25 order 42 generated the hypothesis. Contrasts meeting the rule (separate Holm family, order 42): PlantDoc vs B12, PlantVillage vs B12, iNatAg-100 vs B12.

Table 7. CLOVE-D12 contrasts pooled over all completed class orders (orders crossed in the bootstrap; exploratory replication of the order-42 confirmatory test; the same source group can appear under several orders, so McNemar counts are not independent).

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| Bean Leaf | D12 − B2 | open | -22.2 | [-61.1, +0.0] | 0.524 | 1.000 | 0.219 | 1/5 | 18 |
| Bean Leaf | D12 − B1 | open | -5.6 | [-44.4, +27.8] | 0.872 | 1.000 | 1.000 | 4/5 | 18 |
| DeepWeeds | D12 − B2 | open | -14.8 | [-24.1, -5.6] | 0.001 | 0.027 | 0.021 | 1/9 | 54 |
| DeepWeeds | D12 − B1 | open | -13.0 | [-22.2, -3.7] | 0.005 | 0.125 | 0.065 | 2/9 | 54 |
| FUSARIUM 22 | D12 − B2 | open | +8.3 | [-16.7, +33.3] | 0.743 | 1.000 | 1.000 | 4/3 | 12 |
| FUSARIUM 22 | D12 − B1 | open | -16.7 | [-50.0, +16.7] | 0.430 | 1.000 | 0.727 | 3/5 | 12 |
| Dangerous Insects | D12 − B2 | open | -10.6 | [-21.2, -0.0] | 0.036 | 0.805 | 0.375 | 1/4 | 30 |
| Dangerous Insects | D12 − B1 | open | -6.1 | [-21.2, +9.1] | 0.523 | 1.000 | 0.727 | 3/5 | 30 |
| IP102 (25 cl.) | D12 − B2 | open | +16.7 | [+8.3, +26.1] | 0.001 | 0.014 | 0.000 | 37/7 | 180 |
| IP102 (25 cl.) | D12 − B1 | open | +6.1 | [-7.2, +21.1] | 0.438 | 1.000 | 0.108 | 25/14 | 180 |
| SBRD | D12 − B2 | open | -8.3 | [-27.8, +8.3] | 0.466 | 1.000 | 0.581 | 5/8 | 36 |
| SBRD | D12 − B1 | open | -13.9 | [-27.8, -0.0] | 0.028 | 0.672 | 0.180 | 2/7 | 36 |
| Soybean Diseases (7 cl.) | D12 − B2 | open | +5.6 | [-13.9, +22.2] | 0.590 | 1.000 | 0.791 | 8/6 | 36 |
| Soybean Diseases (7 cl.) | D12 − B1 | open | +2.8 | [-13.9, +19.4] | 0.693 | 1.000 | 1.000 | 7/6 | 36 |
| Soybean Seeds | D12 − B2 | open | -12.5 | [-33.3, +8.3] | 0.345 | 1.000 | 0.375 | 1/4 | 24 |
| Soybean Seeds | D12 − B1 | open | -12.5 | [-29.2, +0.0] | 0.078 | 1.000 | 0.453 | 2/5 | 24 |
| Yellow Rust 19 | D12 − B2 | open | -13.3 | [-40.0, +13.3] | 0.227 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | open | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | open | +5.6 | [-1.1, +13.3] | 0.152 | 1.000 | 0.229 | 33/23 | 180 |
| iNatAg-25 | D12 − B1 | open | +2.8 | [-5.6, +11.1] | 0.572 | 1.000 | 0.551 | 25/20 | 180 |
| SAGE Corn | D12 − B2 | open | +3.2 | [-6.3, +12.7] | 0.597 | 1.000 | 0.832 | 12/10 | 61 |
| SAGE Corn | D12 − B1 | open | +6.3 | [-3.2, +14.3] | 0.196 | 1.000 | 0.503 | 12/8 | 61 |
| SAGE Soybean | D12 − B2 | open | +1.8 | [-5.3, +8.8] | 0.706 | 1.000 | 1.000 | 4/3 | 57 |
| SAGE Soybean | D12 − B1 | open | +10.5 | [+5.3, +17.5] | 0.002 | 0.052 | 0.031 | 6/0 | 57 |
| SAGE Sugarcane | D12 − B2 | open | -4.2 | [-20.8, +16.7] | 0.792 | 1.000 | 1.000 | 4/5 | 24 |
| SAGE Sugarcane | D12 − B1 | open | -12.5 | [-25.0, -0.0] | 0.035 | 0.805 | 0.508 | 3/6 | 24 |
| SAGE Tomato | D12 − B2 | open | -11.1 | [-22.2, +0.0] | 0.069 | 1.000 | 0.289 | 2/6 | 36 |
| SAGE Tomato | D12 − B1 | open | -5.6 | [-16.7, +5.6] | 0.451 | 1.000 | 0.688 | 2/4 | 36 |
| Bean Leaf | D12 − B2 | forced | -22.2 | [-61.1, +0.0] | 0.524 | 1.000 | 0.219 | 1/5 | 18 |
| Bean Leaf | D12 − B1 | forced | -5.6 | [-44.4, +27.8] | 0.872 | 1.000 | 1.000 | 4/5 | 18 |
| DeepWeeds | D12 − B2 | forced | +3.7 | [-5.6, +14.8] | 0.562 | 1.000 | 0.774 | 7/5 | 54 |
| DeepWeeds | D12 − B1 | forced | -1.9 | [-18.5, +11.1] | 0.922 | 1.000 | 1.000 | 5/6 | 54 |
| FUSARIUM 22 | D12 − B2 | forced | +16.7 | [-8.3, +41.7] | 0.290 | 1.000 | 0.688 | 4/2 | 12 |
| FUSARIUM 22 | D12 − B1 | forced | -8.3 | [-41.7, +25.0] | 0.788 | 1.000 | 1.000 | 3/4 | 12 |
| Dangerous Insects | D12 − B2 | forced | -10.6 | [-24.2, +1.5] | 0.085 | 1.000 | 0.453 | 2/5 | 30 |
| Dangerous Insects | D12 − B1 | forced | -9.1 | [-24.2, +3.0] | 0.191 | 1.000 | 0.508 | 3/6 | 30 |
| IP102 (25 cl.) | D12 − B2 | forced | +32.2 | [+23.3, +41.7] | 0.001 | 0.014 | 0.000 | 66/8 | 180 |
| IP102 (25 cl.) | D12 − B1 | forced | +22.2 | [+10.6, +35.6] | 0.001 | 0.014 | 0.000 | 49/9 | 180 |
| SBRD | D12 − B2 | forced | +5.6 | [-19.4, +27.8] | 0.708 | 1.000 | 0.774 | 7/5 | 36 |
| SBRD | D12 − B1 | forced | -5.6 | [-22.2, +11.1] | 0.549 | 1.000 | 0.754 | 4/6 | 36 |
| Soybean Diseases (7 cl.) | D12 − B2 | forced | +16.7 | [-0.1, +33.3] | 0.051 | 1.000 | 0.238 | 12/6 | 36 |
| Soybean Diseases (7 cl.) | D12 − B1 | forced | +11.1 | [-2.8, +25.0] | 0.081 | 1.000 | 0.454 | 10/6 | 36 |
| Soybean Seeds | D12 − B2 | forced | -4.2 | [-25.0, +12.6] | 0.872 | 1.000 | 1.000 | 1/2 | 24 |
| Soybean Seeds | D12 − B1 | forced | -12.5 | [-29.2, +0.0] | 0.155 | 1.000 | 0.453 | 2/5 | 24 |
| Yellow Rust 19 | D12 − B2 | forced | -13.3 | [-40.0, +13.3] | 0.227 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | forced | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | forced | +10.0 | [+5.0, +15.6] | 0.001 | 0.014 | 0.025 | 38/20 | 180 |
| iNatAg-25 | D12 − B1 | forced | +6.1 | [-1.1, +12.8] | 0.093 | 1.000 | 0.169 | 32/21 | 180 |
| SAGE Corn | D12 − B2 | forced | +11.1 | [-0.0, +22.2] | 0.067 | 1.000 | 0.230 | 16/9 | 61 |
| SAGE Corn | D12 − B1 | forced | +14.3 | [+4.8, +25.4] | 0.012 | 0.288 | 0.108 | 17/8 | 61 |
| SAGE Soybean | D12 − B2 | forced | +8.8 | [+1.8, +15.8] | 0.021 | 0.483 | 0.125 | 6/1 | 57 |
| SAGE Soybean | D12 − B1 | forced | +12.3 | [+5.3, +19.3] | 0.001 | 0.025 | 0.016 | 7/0 | 57 |
| SAGE Sugarcane | D12 − B2 | forced | +12.5 | [-8.3, +33.3] | 0.327 | 1.000 | 0.549 | 7/4 | 24 |
| SAGE Sugarcane | D12 − B1 | forced | +4.2 | [-12.5, +20.8] | 0.782 | 1.000 | 1.000 | 5/4 | 24 |
| SAGE Tomato | D12 − B2 | forced | -0.0 | [-13.9, +13.9] | 1.000 | 1.000 | 1.000 | 5/5 | 36 |
| SAGE Tomato | D12 − B1 | forced | +0.0 | [-13.9, +13.9] | 1.000 | 1.000 | 1.000 | 6/6 | 36 |
| Bean Leaf | D12 − B2 | calibrated | -22.2 | [-61.1, +0.0] | 0.524 | 1.000 | 0.219 | 1/5 | 18 |
| Bean Leaf | D12 − B1 | calibrated | -5.6 | [-44.4, +27.8] | 0.872 | 1.000 | 1.000 | 4/5 | 18 |
| DeepWeeds | D12 − B2 | calibrated | -3.7 | [-13.0, +7.4] | 0.454 | 1.000 | 0.754 | 4/6 | 54 |
| DeepWeeds | D12 − B1 | calibrated | -5.6 | [-24.1, +9.3] | 0.570 | 1.000 | 0.549 | 4/7 | 54 |
| FUSARIUM 22 | D12 − B2 | calibrated | +8.3 | [-16.7, +33.3] | 0.743 | 1.000 | 1.000 | 4/3 | 12 |
| FUSARIUM 22 | D12 − B1 | calibrated | -8.3 | [-33.3, +16.7] | 0.717 | 1.000 | 1.000 | 3/4 | 12 |
| Dangerous Insects | D12 − B2 | calibrated | -7.6 | [-18.2, +3.0] | 0.182 | 1.000 | 0.688 | 2/4 | 30 |
| Dangerous Insects | D12 − B1 | calibrated | -3.0 | [-18.2, +12.1] | 0.805 | 1.000 | 1.000 | 4/5 | 30 |
| IP102 (25 cl.) | D12 − B2 | calibrated | +26.7 | [+17.2, +37.8] | 0.001 | 0.014 | 0.000 | 56/8 | 180 |
| IP102 (25 cl.) | D12 − B1 | calibrated | +16.7 | [+4.4, +32.2] | 0.004 | 0.100 | 0.000 | 41/11 | 180 |
| SBRD | D12 − B2 | calibrated | +0.0 | [-19.4, +19.4] | 1.000 | 1.000 | 1.000 | 6/6 | 36 |
| SBRD | D12 − B1 | calibrated | -11.1 | [-25.0, +2.8] | 0.120 | 1.000 | 0.344 | 3/7 | 36 |
| Soybean Diseases (7 cl.) | D12 − B2 | calibrated | +13.9 | [-2.8, +30.6] | 0.055 | 1.000 | 0.332 | 11/6 | 36 |
| Soybean Diseases (7 cl.) | D12 − B1 | calibrated | +8.3 | [-2.8, +22.2] | 0.154 | 1.000 | 0.607 | 9/6 | 36 |
| Soybean Seeds | D12 − B2 | calibrated | -16.7 | [-33.3, +0.0] | 0.055 | 1.000 | 0.125 | 0/4 | 24 |
| Soybean Seeds | D12 − B1 | calibrated | -25.0 | [-37.5, -12.5] | 0.001 | 0.014 | 0.070 | 1/7 | 24 |
| Yellow Rust 19 | D12 − B2 | calibrated | -13.3 | [-40.0, +13.3] | 0.318 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | calibrated | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | calibrated | +10.0 | [+5.0, +15.6] | 0.001 | 0.014 | 0.025 | 38/20 | 180 |
| iNatAg-25 | D12 − B1 | calibrated | +7.8 | [-0.0, +15.0] | 0.056 | 1.000 | 0.065 | 32/18 | 180 |
| SAGE Corn | D12 − B2 | calibrated | +11.1 | [-0.0, +22.2] | 0.067 | 1.000 | 0.230 | 16/9 | 61 |
| SAGE Corn | D12 − B1 | calibrated | +14.3 | [+4.8, +25.4] | 0.012 | 0.276 | 0.108 | 17/8 | 61 |

Table 8. Data sets with ≥25 classes: contrasts pooled over completed class orders (exploratory).

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| IP102 (25 cl.) | D12 − B2 | open | +16.7 | [+8.3, +26.1] | 0.001 | 0.003 | 0.000 | 37/7 | 180 |
| IP102 (96 cl.) | D12 − B12 | open | +4.5 | [+1.7, +7.9] | 0.003 | 0.006 | 0.057 | 11/3 | 177 |
| iNatAg-100 | D12 − B12 | open | +9.0 | [+5.6, +12.4] | 0.001 | 0.003 | 0.002 | 20/4 | 177 |
| iNatAg-25 | D12 − B2 | open | +5.6 | [-1.1, +13.3] | 0.152 | 0.152 | 0.229 | 33/23 | 180 |
| PlantDoc | D12 − B12 | open | +27.3 | [+18.2, +36.4] | 0.001 | 0.003 | 0.000 | 21/3 | 66 |
| PlantVillage | D12 − B12 | open | +26.7 | [+17.8, +35.6] | 0.001 | 0.003 | 0.000 | 32/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | forced | +32.2 | [+23.3, +41.7] | 0.001 | 0.003 | 0.000 | 66/8 | 180 |
| IP102 (96 cl.) | D12 − B12 | forced | +5.6 | [+2.3, +9.0] | 0.002 | 0.003 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | forced | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | forced | +10.0 | [+5.0, +15.6] | 0.001 | 0.003 | 0.025 | 38/20 | 180 |
| PlantDoc | D12 − B12 | forced | +33.3 | [+25.8, +40.9] | 0.001 | 0.003 | 0.000 | 23/1 | 66 |
| PlantVillage | D12 − B12 | forced | +30.0 | [+21.1, +38.9] | 0.001 | 0.003 | 0.000 | 35/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | calibrated | +26.7 | [+17.2, +37.8] | 0.001 | 0.003 | 0.000 | 56/8 | 180 |
| IP102 (96 cl.) | D12 − B12 | calibrated | +5.6 | [+2.3, +9.0] | 0.002 | 0.003 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | calibrated | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | calibrated | +10.0 | [+5.0, +15.6] | 0.001 | 0.003 | 0.025 | 38/20 | 180 |
| PlantDoc | D12 − B12 | calibrated | +31.8 | [+24.2, +40.9] | 0.001 | 0.003 | 0.000 | 22/1 | 66 |
| PlantVillage | D12 − B12 | calibrated | +27.8 | [+18.9, +36.7] | 0.001 | 0.003 | 0.000 | 35/10 | 90 |

Table 9. Pre-registered confirmation on never-examined image groups, final session (columns as in Table 5).

| Arm | BA | $BA_f$ | Old | New | Rej. | FR | AUROC | Calls |
|---|---|---|---|---|---|---|---|---|
| DeepWeeds, final session 2, 18 known / 6 held-out queries | | | | | | | | |
| C1 indep. verification (CGV) | 17 | 17 | 13 | 33 | 67 | 61 | 0.62 | 7.0 |
| B12 retrieve-then-verify | 6 | 39 | 7 | 0 | 100 | 89 | 0.69 | 1.0 |
| B0 zero-shot | 28 | 33 | 27 | 33 | 33 | 22 | 0.61 | 1.0 |
| B1 five-shot ICL | 33 | 39 | 40 | 0 | 17 | 17 | 0.58 | 1.0 |
| B2 uniform memory | 17 | 39 | 20 | 0 | 33 | 61 | 0.59 | 1.0 |
| index-only top-1 (reference, not Qwen) | 39 | – | – | – | – | – | – | 0 |
| Dangerous Insects, final session 2, 33 known / 12 held-out queries | | | | | | | | |
| C1 indep. verification (CGV) | 67 | 70 | 79 | 33 | 42 | 12 | 0.71 | 12.0 |
| B12 retrieve-then-verify | 33 | 39 | 37 | 22 | 100 | 39 | 0.81 | 1.0 |
| B0 zero-shot | 67 | 67 | 75 | 44 | 33 | 0 | 0.85 | 1.0 |
| B1 five-shot ICL | 55 | 58 | 62 | 33 | 17 | 9 | 0.84 | 1.0 |
| B2 uniform memory | 61 | 64 | 67 | 44 | 50 | 9 | 0.85 | 1.0 |
| index-only top-1 (reference, not Qwen) | 33 | – | – | – | – | – | – | 0 |

Table 10. Pre-registered paired contrasts on the confirmatory cohorts (points). The confirmatory family is the open-world rows; forced-choice rows are secondary. Holm correction within each metric over 3 contrasts × 2 data sets.

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar $p$ | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| DeepWeeds | C1 − B2 | open | +0.0 | [-22.2, +16.7] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| DeepWeeds | B12 − B2 | open | -11.1 | [-16.7, +0.0] | 0.084 | 0.405 | 0.500 | 0/2 | 18 |
| DeepWeeds | B1 − B2 | open | +16.7 | [+0.0, +33.3] | 0.081 | 0.405 | 0.453 | 5/2 | 18 |
| Dangerous Insects | C1 − B2 | open | +6.1 | [-6.1, +18.2] | 0.426 | 0.852 | 0.754 | 6/4 | 33 |
| Dangerous Insects | B12 − B2 | open | -27.3 | [-39.4, -15.2] | 0.001 | 0.003 | 0.022 | 2/11 | 33 |
| Dangerous Insects | B1 − B2 | open | -6.1 | [-15.2, +3.0] | 0.274 | 0.822 | 0.625 | 1/3 | 33 |
| DeepWeeds | C1 − B2 | forced | -22.2 | [-44.4, +0.0] | 0.058 | 0.290 | 0.289 | 2/6 | 18 |
| DeepWeeds | B12 − B2 | forced | +0.0 | [-16.7, +16.7] | 1.000 | 1.000 | 1.000 | 3/3 | 18 |
| DeepWeeds | B1 − B2 | forced | +0.0 | [-11.1, +11.1] | 1.000 | 1.000 | 1.000 | 1/1 | 18 |
| Dangerous Insects | C1 − B2 | forced | +6.1 | [-9.1, +18.2] | 0.484 | 1.000 | 0.754 | 6/4 | 33 |
| Dangerous Insects | B12 − B2 | forced | -24.2 | [-36.4, -12.1] | 0.001 | 0.003 | 0.039 | 2/10 | 33 |
| Dangerous Insects | B1 − B2 | forced | -6.1 | [-15.2, +3.0] | 0.274 | 1.000 | 0.625 | 1/3 | 33 |

Table 11. Model calls, prompt tokens, images per prompt and seconds per query on the confirmatory cohorts.

| Arm | Records | Calls | Tokens | Images | s (mean) | s (p95) |
|---|---|---|---|---|---|---|
| B0 zero-shot | 185.0 | 1.0 | 364 | 1.0 | 2.0 | 2.4 |
| B1 five-shot ICL | 184.0 | 1.0 | 1338 | 14.7 | 5.5 | 8.3 |
| B12 retrieve-then-verify | 184.0 | 1.0 | 730 | 5.1 | 2.8 | 3.5 |
| B2 uniform memory | 184.0 | 1.0 | 1194 | 9.1 | 4.3 | 5.5 |
| C1 indep. verification (CGV) | 184.0 | 7.8 | 501 | 1.0 | 14.5 | 22.1 |

Table 12. Round 5 development (class order 42, final session; DeepWeeds and Mango three, Insects two queries per class). BA open / forced / calibrated (%; calibration in-sample on development, descriptive only); old/new; Rej. and $\text{Rej.}_{\text{cal}}$: held-out answered UNKNOWN; AUROC; calls.

| Arm | BA | $\text{BA}_f$ | $\text{BA}_{\text{cal}}$ | Old | New | Rej. | $\text{Rej.}_{\text{cal}}$ | AUROC | Calls |
|---|---|---|---|---|---|---|---|---|---|
| Mango Leaf | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 67 | 72 | 72 | 60 | 100 | 17 | 0 | 0.81 | 1.0 |
| B2 uniform memory | 67 | 67 | 67 | 60 | 100 | 17 | 0 | 0.83 | 1.0 |
| D2 dense memory | 56 | 61 | 56 | 47 | 100 | 17 | 0 | 0.76 | 1.0 |
| D3 dense + fixed orch. | 61 | 72 | 67 | 53 | 100 | 0 | 0 | 0.63 | 2.0 |
| D4 dense + autonomous orch. | 61 | 61 | 61 | 53 | 100 | 50 | 17 | 0.86 | 3.7 |
| D12 dense + retrieve-then-verify | 61 | 78 | 67 | 53 | 100 | 50 | 0 | 0.67 | 1.0 |
| DeepWeeds | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 39 | 39 | 39 | 47 | 0 | 17 | 0 | 0.52 | 1.0 |
| B2 uniform memory | 28 | 44 | 39 | 27 | 33 | 67 | 17 | 0.69 | 1.0 |
| D2 dense memory | 22 | 39 | 33 | 27 | 0 | 50 | 33 | 0.62 | 1.0 |
| D3 dense + fixed orch. | 22 | 22 | 22 | 27 | 0 | 67 | 17 | 0.65 | 2.0 |
| D4 dense + autonomous orch. | 17 | 22 | 22 | 20 | 0 | 50 | 0 | 0.61 | 3.8 |
| D12 dense + retrieve-then-verify | 39 | 61 | 56 | 40 | 33 | 67 | 67 | 0.69 | 1.0 |
| Dangerous Insects | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 55 | 55 | 55 | 69 | 17 | 62 | 62 | 0.95 | 1.0 |
| B2 uniform memory | 64 | 64 | 64 | 81 | 17 | 88 | 75 | 0.98 | 1.0 |
| D2 dense memory | 55 | 55 | 55 | 75 | 0 | 88 | 75 | 0.99 | 1.0 |
| D3 dense + fixed orch. | 64 | 73 | 68 | 75 | 33 | 88 | 88 | 0.98 | 2.0 |
| D4 dense + autonomous orch. | 73 | 73 | 73 | 81 | 50 | 75 | 88 | 0.98 | 3.3 |
| D12 dense + retrieve-then-verify | 59 | 68 | 64 | 69 | 33 | 88 | 88 | 0.98 | 1.0 |
| IDC | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 25 | 25 | 25 | 0 | 100 | — | — | — | 1.0 |
| B2 uniform memory | 25 | 33 | 33 | 22 | 33 | — | — | — | 1.0 |
| D12 dense + retrieve-then-verify | 33 | 50 | 42 | 33 | 33 | — | — | — | 1.0 |

Table 13. Round 5 paired development contrasts (exploratory).

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| DeepWeeds | D2 − B2 | open | -5.6 | [-16.7, +0.0] | 0.565 | 1.000 | 1.000 | 0/1 | 18 |
| DeepWeeds | D3 − B2 | open | -5.6 | [-22.2, +11.1] | 0.717 | 1.000 | 1.000 | 1/2 | 18 |
| DeepWeeds | D4 − B2 | open | -11.1 | [-22.2, +0.0] | 0.160 | 1.000 | 0.500 | 0/2 | 18 |
| DeepWeeds | D12 − B2 | open | +11.1 | [-5.6, +27.8] | 0.386 | 1.000 | 0.625 | 3/1 | 18 |
| DeepWeeds | D2 − B1 | open | -16.7 | [-27.8, -5.6] | 0.019 | 0.380 | 0.250 | 0/3 | 18 |
| DeepWeeds | B1 − B2 | open | +11.1 | [-5.6, +27.8] | 0.115 | 1.000 | 0.625 | 3/1 | 18 |
| IDC | D12 − B2 | open | +8.3 | [-16.7, +33.3] | 0.700 | 1.000 | 1.000 | 3/2 | 12 |
| IDC | B1 − B2 | open | +0.0 | [-16.7, +16.7] | 0.826 | 1.000 | 1.000 | 2/2 | 12 |
| Dangerous Insects | D2 − B2 | open | -9.1 | [-18.2, +0.0] | 0.116 | 1.000 | 0.500 | 0/2 | 22 |
| Dangerous Insects | D3 − B2 | open | +0.0 | [-9.1, +9.1] | 1.000 | 1.000 | 1.000 | 1/1 | 22 |
| Dangerous Insects | D4 − B2 | open | +9.1 | [-4.5, +22.7] | 0.385 | 1.000 | 0.688 | 4/2 | 22 |
| Dangerous Insects | D12 − B2 | open | -4.5 | [-22.7, +9.1] | 0.784 | 1.000 | 1.000 | 2/3 | 22 |
| Dangerous Insects | D2 − B1 | open | +0.0 | [-9.1, +9.1] | 1.000 | 1.000 | 1.000 | 1/1 | 22 |
| Dangerous Insects | B1 − B2 | open | -9.1 | [-22.7, +4.5] | 0.298 | 1.000 | 0.625 | 1/3 | 22 |
| Mango Leaf | D2 − B2 | open | -11.1 | [-22.2, +0.0] | 0.176 | 1.000 | 0.500 | 0/2 | 18 |
| Mango Leaf | D3 − B2 | open | -5.6 | [-22.2, +11.1] | 0.688 | 1.000 | 1.000 | 2/3 | 18 |
| Mango Leaf | D4 − B2 | open | -5.6 | [-16.7, +0.0] | 0.566 | 1.000 | 1.000 | 0/1 | 18 |
| Mango Leaf | D12 − B2 | open | -5.6 | [-22.2, +16.7] | 0.636 | 1.000 | 1.000 | 2/3 | 18 |
| Mango Leaf | D2 − B1 | open | -11.1 | [-16.7, +0.0] | 0.077 | 1.000 | 0.500 | 0/2 | 18 |
| Mango Leaf | B1 − B2 | open | +0.0 | [-11.1, +11.1] | 1.000 | 1.000 | 1.000 | 1/1 | 18 |
| DeepWeeds | D2 − B2 | forced | -5.6 | [-16.7, +0.0] | 0.565 | 1.000 | 1.000 | 0/1 | 18 |
| DeepWeeds | D3 − B2 | forced | -22.2 | [-33.3, -16.7] | 0.001 | 0.010 | 0.125 | 0/4 | 18 |
| DeepWeeds | D4 − B2 | forced | -22.2 | [-33.3, -16.7] | 0.001 | 0.010 | 0.125 | 0/4 | 18 |
| DeepWeeds | D12 − B2 | forced | +16.7 | [-0.1, +33.3] | 0.167 | 1.000 | 0.375 | 4/1 | 18 |
| DeepWeeds | D2 − B1 | forced | +0.0 | [+0.0, +0.0] | 1.000 | 1.000 | 1.000 | 0/0 | 18 |
| DeepWeeds | B1 − B2 | forced | -5.6 | [-16.7, +0.0] | 0.565 | 1.000 | 1.000 | 0/1 | 18 |
| IDC | D12 − B2 | forced | +16.7 | [-8.3, +50.0] | 0.489 | 1.000 | 0.688 | 4/2 | 12 |
| IDC | B1 − B2 | forced | -8.3 | [-25.0, +8.3] | 0.639 | 1.000 | 1.000 | 1/2 | 12 |
| Dangerous Insects | D2 − B2 | forced | -9.1 | [-18.2, +0.0] | 0.116 | 1.000 | 0.500 | 0/2 | 22 |
| Dangerous Insects | D3 − B2 | forced | +9.1 | [+0.0, +18.2] | 0.118 | 1.000 | 0.625 | 3/1 | 22 |
| Dangerous Insects | D4 − B2 | forced | +9.1 | [-4.5, +22.7] | 0.385 | 1.000 | 0.688 | 4/2 | 22 |
| Dangerous Insects | D12 − B2 | forced | +4.5 | [-13.6, +18.2] | 0.834 | 1.000 | 1.000 | 4/3 | 22 |
| Dangerous Insects | D2 − B1 | forced | +0.0 | [-9.1, +9.1] | 1.000 | 1.000 | 1.000 | 1/1 | 22 |
| Dangerous Insects | B1 − B2 | forced | -9.1 | [-22.7, +4.5] | 0.298 | 1.000 | 0.625 | 1/3 | 22 |
| Mango Leaf | D2 − B2 | forced | -5.6 | [-16.7, +0.0] | 0.566 | 1.000 | 1.000 | 0/1 | 18 |
| Mango Leaf | D3 − B2 | forced | +5.6 | [-5.6, +16.7] | 0.413 | 1.000 | 1.000 | 2/1 | 18 |
| Mango Leaf | D4 − B2 | forced | -5.6 | [-16.7, +0.0] | 0.566 | 1.000 | 1.000 | 0/1 | 18 |
| Mango Leaf | D12 − B2 | forced | +11.1 | [-5.6, +27.8] | 0.295 | 1.000 | 0.625 | 3/1 | 18 |
| Mango Leaf | D2 − B1 | forced | -11.1 | [-16.7, +0.0] | 0.077 | 1.000 | 0.500 | 0/2 | 18 |
| Mango Leaf | B1 − B2 | forced | +5.6 | [+0.0, +16.7] | 0.599 | 1.000 | 1.000 | 1/0 | 18 |
| DeepWeeds | D2 − B2 | calibrated | -5.6 | [-16.7, +0.0] | 0.565 | 1.000 | 1.000 | 0/1 | 18 |
| DeepWeeds | D3 − B2 | calibrated | -16.7 | [-27.8, -5.6] | 0.020 | 0.400 | 0.375 | 1/4 | 18 |
| DeepWeeds | D4 − B2 | calibrated | -16.7 | [-27.8, -5.6] | 0.020 | 0.400 | 0.375 | 1/4 | 18 |
| DeepWeeds | D12 − B2 | calibrated | +16.7 | [+0.0, +33.3] | 0.182 | 1.000 | 0.375 | 4/1 | 18 |
| DeepWeeds | D2 − B1 | calibrated | -5.6 | [-16.7, +0.0] | 0.602 | 1.000 | 1.000 | 0/1 | 18 |
| DeepWeeds | B1 − B2 | calibrated | +0.0 | [-11.1, +11.1] | 1.000 | 1.000 | 1.000 | 1/1 | 18 |
| IDC | D12 − B2 | calibrated | +8.3 | [-25.0, +41.7] | 0.826 | 1.000 | 1.000 | 4/3 | 12 |
| IDC | B1 − B2 | calibrated | -8.3 | [-25.0, +8.3] | 0.639 | 1.000 | 1.000 | 1/2 | 12 |
| Dangerous Insects | D2 − B2 | calibrated | -9.1 | [-18.2, +0.0] | 0.116 | 1.000 | 0.500 | 0/2 | 22 |
| Dangerous Insects | D3 − B2 | calibrated | +4.5 | [+0.0, +9.1] | 0.460 | 1.000 | 1.000 | 2/1 | 22 |
| Dangerous Insects | D4 − B2 | calibrated | +9.1 | [-4.5, +22.7] | 0.385 | 1.000 | 0.688 | 4/2 | 22 |
| Dangerous Insects | D12 − B2 | calibrated | +0.0 | [-18.2, +18.2] | 1.000 | 1.000 | 1.000 | 3/3 | 22 |
| Dangerous Insects | D2 − B1 | calibrated | +0.0 | [-9.1, +9.1] | 1.000 | 1.000 | 1.000 | 1/1 | 22 |
| Dangerous Insects | B1 − B2 | calibrated | -9.1 | [-22.7, +4.5] | 0.298 | 1.000 | 0.625 | 1/3 | 22 |
| Mango Leaf | D2 − B2 | calibrated | -11.1 | [-22.2, +0.0] | 0.176 | 1.000 | 0.500 | 0/2 | 18 |
| Mango Leaf | D3 − B2 | calibrated | +0.0 | [-16.7, +16.7] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| Mango Leaf | D4 − B2 | calibrated | -5.6 | [-16.7, +0.0] | 0.566 | 1.000 | 1.000 | 0/1 | 18 |
| Mango Leaf | D12 − B2 | calibrated | +0.0 | [-16.8, +16.8] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| Mango Leaf | D2 − B1 | calibrated | -16.7 | [-27.8, -5.6] | 0.028 | 0.504 | 0.250 | 0/3 | 18 |
| Mango Leaf | B1 − B2 | calibrated | +5.6 | [+0.0, +16.7] | 0.599 | 1.000 | 1.000 | 1/0 | 18 |

Table 14. Round 5 confirmation on never-examined groups (final session, mean over completed class orders).

| Arm | BA | $BA_f$ | $BA_{cal}$ | Old | New | Rej. | $Rej._{cal}$ | AUROC | Calls |
|---|---|---|---|---|---|---|---|---|---|
| DeepWeeds | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 39 | 50 | 41 | 38 | 44 | 39 | 11 | 0.60 | 1.0 |
| B2 uniform memory | 41 | 44 | 38 | 40 | 44 | 39 | 28 | 0.64 | 1.0 |
| D12 dense + retrieve-then-verify | 26 | 48 | 36 | 27 | 22 | 72 | 56 | 0.71 | 1.0 |
| Dangerous Insects | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 61 | 67 | 61 | 67 | 44 | 67 | 58 | 0.83 | 1.0 |
| B2 uniform memory | 65 | 68 | 65 | 67 | 61 | 75 | 75 | 0.87 | 1.0 |
| D12 dense + retrieve-then-verify | 55 | 58 | 58 | 58 | 44 | 92 | 92 | 0.83 | 1.0 |
| SBRD | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 58 | 64 | 59 | 50 | 100 | 17 | 0 | 0.58 | 1.0 |
| B2 uniform memory | 53 | 53 | 50 | 53 | 50 | 0 | 0 | 0.52 | 1.0 |
| D12 dense + retrieve-then-verify | 44 | 58 | 52 | 40 | 67 | 67 | 39 | 0.77 | 1.0 |
| Soybean Seeds | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 71 | 79 | 80 | 67 | 83 | 50 | 0 | 0.78 | 1.0 |
| B2 uniform memory | 71 | 71 | 77 | 61 | 100 | 0 | 0 | 0.62 | 1.0 |
| D12 dense + retrieve-then-verify | 58 | 67 | 60 | 50 | 83 | 33 | 33 | 0.78 | 1.0 |
| Bean Leaf | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 56 | 56 | 56 | 58 | 50 | — | — | — | 1.0 |
| B2 uniform memory | 72 | 72 | 72 | 83 | 50 | — | — | — | 1.0 |
| D12 dense + retrieve-then-verify | 50 | 50 | 50 | 75 | 0 | — | — | — | 1.0 |
| iNatAg-25 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 53 | 56 | 56 | 54 | 49 | 36 | 4 | 0.67 | 1.0 |
| B2 uniform memory | 50 | 52 | 53 | 56 | 31 | 44 | 4 | 0.70 | 1.0 |
| D12 dense + retrieve-then-verify | 56 | 62 | 64 | 56 | 53 | 58 | 7 | 0.67 | 1.0 |
| IP102 (25 cl.) | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 25 | 26 | 26 | 24 | 29 | 49 | 18 | 0.66 | 1.0 |
| B2 uniform memory | 14 | 16 | 15 | 16 | 11 | 67 | 16 | 0.63 | 1.0 |
| D12 dense + retrieve-then-verify | 31 | 48 | 43 | 27 | 44 | 69 | 36 | 0.66 | 1.0 |
| Soybean Diseases (7 cl.) | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 36 | 39 | 38 | 33 | 50 | 0 | 0 | 0.74 | 1.0 |
| B2 uniform memory | 33 | 33 | 33 | 30 | 50 | 0 | 0 | 0.89 | 1.0 |
| D12 dense + retrieve-then-verify | 39 | 50 | 43 | 30 | 83 | 17 | 0 | 0.54 | 1.0 |
| iNatAg-100 | | | | | | | | | |
| D12 dense + retrieve-then-verify | 15 | 16 | 16 | 15 | 14 | 65 | 8 | 0.62 | 1.0 |
| B12 uniform + retrieve-then-verify | 6 | 6 | 6 | 6 | 0 | 73 | 4 | 0.46 | 1.0 |
| PlantDoc | | | | | | | | | |
| D12 dense + retrieve-then-verify | 47 | 53 | 52 | 44 | 58 | 20 | 7 | 0.56 | 1.0 |
| B12 uniform + retrieve-then-verify | 20 | 20 | 20 | 24 | 0 | 27 | 0 | 0.48 | 1.0 |
| PlantVillage | | | | | | | | | |
| D12 dense + retrieve-then-verify | 38 | 46 | 43 | 36 | 47 | 33 | 0 | 0.45 | 1.0 |
| B12 uniform + retrieve-then-verify | 11 | 16 | 16 | 12 | 7 | 38 | 0 | 0.39 | 1.0 |
| IP102 (96 cl.) | | | | | | | | | |
| D12 dense + retrieve-then-verify | 7 | 10 | 10 | 8 | 5 | 61 | 11 | 0.47 | 1.0 |
| B12 uniform + retrieve-then-verify | 3 | 4 | 4 | 3 | 0 | 72 | 17 | 0.46 | 1.0 |
| Yellow Rust 19 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 40 | 40 | 33 | 50 | 0 | 0 | 0 | 0.84 | 1.0 |
| B2 uniform memory | 47 | 47 | 40 | 58 | 0 | 0 | 0 | 0.82 | 1.0 |
| D12 dense + retrieve-then-verify | 33 | 33 | 27 | 33 | 33 | 100 | 33 | 1.00 | 1.0 |
| FUSARIUM 22 | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 50 | 50 | 42 | 44 | 67 | 0 | 0 | 0.58 | 1.0 |
| B2 uniform memory | 25 | 25 | 25 | 11 | 67 | 0 | 0 | 0.36 | 1.0 |
| D12 dense + retrieve-then-verify | 33 | 42 | 33 | 44 | 0 | 67 | 0 | 0.44 | 1.0 |
| SAGE Corn | | | | | | | | | |
| B1 two-shot ICL (2 of 5 supports shown) | 19 | 19 | 19 | 13 | 56 | 0 | 0 | 0.64 | 1.0 |
| B2 uniform memory | 22 | 22 | 22 | 17 | 56 | 0 | 0 | 0.63 | 1.0 |
| D12 dense + retrieve-then-verify | 25 | 33 | 33 | 28 | 11 | 42 | 17 | 0.70 | 1.0 |

Table 15. Round 5 pre-registered contrasts, class order 42 (open: primary; forced and calibrated: secondary).

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| Bean Leaf | D12 − B2 | open | -44.4 | [-77.8, -11.1] | 0.026 | 0.600 | 0.219 | 1/5 | 9 |
| Bean Leaf | D12 − B1 | open | -33.3 | [-55.6, -11.1] | 0.024 | 0.600 | 0.453 | 2/5 | 9 |
| DeepWeeds | D12 − B2 | open | -16.7 | [-27.8, -5.6] | 0.029 | 0.638 | 0.250 | 0/3 | 18 |
| DeepWeeds | D12 − B1 | open | -16.7 | [-27.8, -5.6] | 0.024 | 0.600 | 0.375 | 1/4 | 18 |
| FUSARIUM 22 | D12 − B2 | open | +8.3 | [-16.7, +33.3] | 0.743 | 1.000 | 1.000 | 4/3 | 12 |
| FUSARIUM 22 | D12 − B1 | open | -16.7 | [-50.0, +16.7] | 0.430 | 1.000 | 0.727 | 3/5 | 12 |
| Dangerous Insects | D12 − B2 | open | -10.6 | [-21.2, -0.0] | 0.036 | 0.735 | 0.375 | 1/4 | 30 |
| Dangerous Insects | D12 − B1 | open | -6.1 | [-21.2, +9.1] | 0.523 | 1.000 | 0.727 | 3/5 | 30 |
| IP102 (25 cl.) | D12 − B2 | open | +25.0 | [+16.7, +33.3] | 0.001 | 0.014 | 0.000 | 16/1 | 60 |
| IP102 (25 cl.) | D12 − B1 | open | +21.7 | [+13.3, +31.7] | 0.001 | 0.014 | 0.001 | 14/1 | 60 |
| SBRD | D12 − B2 | open | +0.0 | [-16.7, +16.7] | 0.962 | 1.000 | 1.000 | 3/3 | 18 |
| SBRD | D12 − B1 | open | -11.1 | [-22.2, +0.0] | 0.166 | 1.000 | 0.500 | 0/2 | 18 |
| Soybean Diseases (7 cl.) | D12 − B2 | open | -5.6 | [-22.2, +11.1] | 0.684 | 1.000 | 1.000 | 4/5 | 18 |
| Soybean Diseases (7 cl.) | D12 − B1 | open | -5.6 | [-22.2, +11.1] | 0.700 | 1.000 | 1.000 | 4/5 | 18 |
| Soybean Seeds | D12 − B2 | open | +0.0 | [-16.7, +16.7] | 1.000 | 1.000 | 1.000 | 1/1 | 12 |
| Soybean Seeds | D12 − B1 | open | -8.3 | [-25.0, +0.0] | 0.580 | 1.000 | 1.000 | 2/3 | 12 |
| Yellow Rust 19 | D12 − B2 | open | -13.3 | [-40.0, +13.3] | 0.227 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | open | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | open | +3.3 | [-6.7, +13.3] | 0.658 | 1.000 | 0.815 | 10/8 | 60 |
| iNatAg-25 | D12 − B1 | open | +5.0 | [-6.7, +16.7] | 0.457 | 1.000 | 0.648 | 11/8 | 60 |
| SAGE Corn | D12 − B2 | open | +3.2 | [-6.3, +12.7] | 0.597 | 1.000 | 0.832 | 12/10 | 61 |
| SAGE Corn | D12 − B1 | open | +6.3 | [-3.2, +14.3] | 0.196 | 1.000 | 0.503 | 12/8 | 61 |
| SAGE Soybean | D12 − B2 | open | +1.8 | [-5.3, +8.8] | 0.706 | 1.000 | 1.000 | 4/3 | 57 |
| SAGE Soybean | D12 − B1 | open | +10.5 | [+5.3, +17.5] | 0.002 | 0.052 | 0.031 | 6/0 | 57 |
| SAGE Sugarcane | D12 − B2 | open | -4.2 | [-20.8, +16.7] | 0.792 | 1.000 | 1.000 | 4/5 | 24 |
| SAGE Sugarcane | D12 − B1 | open | -12.5 | [-25.0, -0.0] | 0.035 | 0.735 | 0.508 | 3/6 | 24 |
| SAGE Tomato | D12 − B2 | open | -11.1 | [-22.2, +0.0] | 0.069 | 1.000 | 0.289 | 2/6 | 36 |
| SAGE Tomato | D12 − B1 | open | -5.6 | [-16.7, +5.6] | 0.451 | 1.000 | 0.688 | 2/4 | 36 |
| Bean Leaf | D12 − B2 | forced | -44.4 | [-77.8, -11.1] | 0.026 | 0.546 | 0.219 | 1/5 | 9 |
| Bean Leaf | D12 − B1 | forced | -33.3 | [-55.6, -11.1] | 0.024 | 0.528 | 0.453 | 2/5 | 9 |
| DeepWeeds | D12 − B2 | forced | +5.6 | [-11.1, +22.2] | 0.760 | 1.000 | 1.000 | 2/1 | 18 |
| DeepWeeds | D12 − B1 | forced | +5.6 | [-11.1, +22.2] | 0.760 | 1.000 | 1.000 | 2/1 | 18 |
| FUSARIUM 22 | D12 − B2 | forced | +16.7 | [-8.3, +41.7] | 0.290 | 1.000 | 0.688 | 4/2 | 12 |
| FUSARIUM 22 | D12 − B1 | forced | -8.3 | [-41.7, +25.0] | 0.788 | 1.000 | 1.000 | 3/4 | 12 |
| Dangerous Insects | D12 − B2 | forced | -10.6 | [-24.2, +1.5] | 0.085 | 1.000 | 0.453 | 2/5 | 30 |
| Dangerous Insects | D12 − B1 | forced | -9.1 | [-24.2, +3.0] | 0.191 | 1.000 | 0.508 | 3/6 | 30 |
| IP102 (25 cl.) | D12 − B2 | forced | +40.0 | [+30.0, +50.0] | 0.001 | 0.014 | 0.000 | 25/1 | 60 |
| IP102 (25 cl.) | D12 − B1 | forced | +35.0 | [+23.3, +46.7] | 0.001 | 0.014 | 0.000 | 23/2 | 60 |
| SBRD | D12 − B2 | forced | +22.2 | [+11.1, +33.3] | 0.004 | 0.100 | 0.219 | 5/1 | 18 |
| SBRD | D12 − B1 | forced | +0.0 | [-22.2, +16.7] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| Soybean Diseases (7 cl.) | D12 − B2 | forced | +5.6 | [-11.1, +22.2] | 0.364 | 1.000 | 1.000 | 6/5 | 18 |
| Soybean Diseases (7 cl.) | D12 − B1 | forced | +5.6 | [-11.1, +22.2] | 0.383 | 1.000 | 1.000 | 6/5 | 18 |
| Soybean Seeds | D12 − B2 | forced | +8.3 | [+0.0, +25.0] | 0.601 | 1.000 | 1.000 | 1/0 | 12 |
| Soybean Seeds | D12 − B1 | forced | -8.3 | [-25.0, +8.3] | 0.664 | 1.000 | 1.000 | 2/3 | 12 |
| Yellow Rust 19 | D12 − B2 | forced | -13.3 | [-40.0, +13.3] | 0.227 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | forced | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | forced | +10.0 | [+1.7, +18.3] | 0.032 | 0.640 | 0.263 | 13/7 | 60 |
| iNatAg-25 | D12 − B1 | forced | +8.3 | [-0.0, +18.3] | 0.099 | 1.000 | 0.359 | 12/7 | 60 |
| SAGE Corn | D12 − B2 | forced | +11.1 | [-0.0, +22.2] | 0.067 | 1.000 | 0.230 | 16/9 | 61 |
| SAGE Corn | D12 − B1 | forced | +14.3 | [+4.8, +25.4] | 0.012 | 0.288 | 0.108 | 17/8 | 61 |
| SAGE Soybean | D12 − B2 | forced | +8.8 | [+1.8, +15.8] | 0.021 | 0.483 | 0.125 | 6/1 | 57 |
| SAGE Soybean | D12 − B1 | forced | +12.3 | [+5.3, +19.3] | 0.001 | 0.026 | 0.016 | 7/0 | 57 |
| SAGE Sugarcane | D12 − B2 | forced | +12.5 | [-8.3, +33.3] | 0.327 | 1.000 | 0.549 | 7/4 | 24 |
| SAGE Sugarcane | D12 − B1 | forced | +4.2 | [-12.5, +20.8] | 0.782 | 1.000 | 1.000 | 5/4 | 24 |
| SAGE Tomato | D12 − B2 | forced | -0.0 | [-13.9, +13.9] | 1.000 | 1.000 | 1.000 | 5/5 | 36 |
| SAGE Tomato | D12 − B1 | forced | +0.0 | [-13.9, +13.9] | 1.000 | 1.000 | 1.000 | 6/6 | 36 |
| Bean Leaf | D12 − B2 | calibrated | -44.4 | [-77.8, -11.1] | 0.026 | 0.572 | 0.219 | 1/5 | 9 |
| Bean Leaf | D12 − B1 | calibrated | -33.3 | [-55.6, -11.1] | 0.024 | 0.552 | 0.453 | 2/5 | 9 |
| DeepWeeds | D12 − B2 | calibrated | +0.0 | [-16.7, +22.2] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| DeepWeeds | D12 − B1 | calibrated | +0.0 | [-16.7, +22.2] | 1.000 | 1.000 | 1.000 | 2/2 | 18 |
| FUSARIUM 22 | D12 − B2 | calibrated | +8.3 | [-16.7, +33.3] | 0.743 | 1.000 | 1.000 | 4/3 | 12 |
| FUSARIUM 22 | D12 − B1 | calibrated | -8.3 | [-33.3, +16.7] | 0.717 | 1.000 | 1.000 | 3/4 | 12 |
| Dangerous Insects | D12 − B2 | calibrated | -7.6 | [-18.2, +3.0] | 0.182 | 1.000 | 0.688 | 2/4 | 30 |
| Dangerous Insects | D12 − B1 | calibrated | -3.0 | [-18.2, +12.1] | 0.805 | 1.000 | 1.000 | 4/5 | 30 |
| IP102 (25 cl.) | D12 − B2 | calibrated | +36.7 | [+26.7, +46.7] | 0.001 | 0.014 | 0.000 | 23/1 | 60 |
| IP102 (25 cl.) | D12 − B1 | calibrated | +31.7 | [+20.0, +43.3] | 0.001 | 0.014 | 0.000 | 21/2 | 60 |
| SBRD | D12 − B2 | calibrated | +11.1 | [-5.6, +27.8] | 0.262 | 1.000 | 0.688 | 4/2 | 18 |
| SBRD | D12 − B1 | calibrated | -11.1 | [-27.8, +5.6] | 0.310 | 1.000 | 0.625 | 1/3 | 18 |
| Soybean Diseases (7 cl.) | D12 − B2 | calibrated | +5.6 | [-11.1, +22.2] | 0.364 | 1.000 | 1.000 | 6/5 | 18 |
| Soybean Diseases (7 cl.) | D12 − B1 | calibrated | +5.6 | [-11.1, +22.2] | 0.383 | 1.000 | 1.000 | 6/5 | 18 |
| Soybean Seeds | D12 − B2 | calibrated | -8.3 | [-25.0, +0.0] | 0.605 | 1.000 | 1.000 | 0/1 | 12 |
| Soybean Seeds | D12 − B1 | calibrated | -25.0 | [-41.7, -8.3] | 0.022 | 0.528 | 0.375 | 1/4 | 12 |
| Yellow Rust 19 | D12 − B2 | calibrated | -13.3 | [-40.0, +13.3] | 0.318 | 1.000 | 0.688 | 2/4 | 15 |
| Yellow Rust 19 | D12 − B1 | calibrated | -6.7 | [-33.3, +13.3] | 0.548 | 1.000 | 1.000 | 2/3 | 15 |
| iNatAg-25 | D12 − B2 | calibrated | +10.0 | [+1.7, +18.3] | 0.032 | 0.672 | 0.263 | 13/7 | 60 |
| iNatAg-25 | D12 − B1 | calibrated | +10.0 | [+1.7, +18.3] | 0.035 | 0.700 | 0.238 | 12/6 | 60 |
| SAGE Corn | D12 − B2 | calibrated | +11.1 | [-0.0, +22.2] | 0.067 | 1.000 | 0.230 | 16/9 | 61 |
| SAGE Corn | D12 − B1 | calibrated | +14.3 | [+4.8, +25.4] | 0.012 | 0.300 | 0.108 | 17/8 | 61 |
| SAGE Soybean | D12 − B2 | calibrated | +7.0 | [-0.0, +15.8] | 0.122 | 1.000 | 0.289 | 6/2 | 57 |
| SAGE Soybean | D12 − B1 | calibrated | +10.5 | [+3.5, +17.5] | 0.009 | 0.234 | 0.070 | 7/1 | 57 |
| SAGE Sugarcane | D12 − B2 | calibrated | +12.5 | [-8.3, +33.3] | 0.327 | 1.000 | 0.549 | 7/4 | 24 |
| SAGE Sugarcane | D12 − B1 | calibrated | +4.2 | [-12.5, +20.8] | 0.782 | 1.000 | 1.000 | 5/4 | 24 |
| SAGE Tomato | D12 − B2 | calibrated | -2.8 | [-13.9, +11.1] | 0.711 | 1.000 | 1.000 | 4/5 | 36 |
| SAGE Tomato | D12 − B1 | calibrated | -2.8 | [-16.7, +11.1] | 0.802 | 1.000 | 1.000 | 5/6 | 36 |

Table 16. Round 5 extension to data sets with ≥25 classes: D12 vs B12 (memory optimisation at equal orchestration) and vs B2, class order 42.

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| IP102 (25 cl.) | D12 − B2 | open | +25.0 | [+16.7, +33.3] | 0.001 | 0.003 | 0.000 | 16/1 | 60 |
| IP102 (96 cl.) | D12 − B12 | open | +4.5 | [+1.7, +7.9] | 0.003 | 0.006 | 0.057 | 11/3 | 177 |
| iNatAg-100 | D12 − B12 | open | +9.0 | [+5.6, +12.4] | 0.001 | 0.003 | 0.002 | 20/4 | 177 |
| iNatAg-25 | D12 − B2 | open | +3.3 | [-6.7, +13.3] | 0.658 | 0.658 | 0.815 | 10/8 | 60 |
| PlantDoc | D12 − B12 | open | +27.3 | [+18.2, +36.4] | 0.001 | 0.003 | 0.000 | 21/3 | 66 |
| PlantVillage | D12 − B12 | open | +26.7 | [+17.8, +35.6] | 0.001 | 0.003 | 0.000 | 32/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | forced | +40.0 | [+30.0, +50.0] | 0.001 | 0.003 | 0.000 | 25/1 | 60 |
| IP102 (96 cl.) | D12 − B12 | forced | +5.6 | [+2.3, +9.0] | 0.002 | 0.004 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | forced | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | forced | +10.0 | [+1.7, +18.3] | 0.032 | 0.032 | 0.263 | 13/7 | 60 |
| PlantDoc | D12 − B12 | forced | +33.3 | [+25.8, +40.9] | 0.001 | 0.003 | 0.000 | 23/1 | 66 |
| PlantVillage | D12 − B12 | forced | +30.0 | [+21.1, +38.9] | 0.001 | 0.003 | 0.000 | 35/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | calibrated | +36.7 | [+26.7, +46.7] | 0.001 | 0.003 | 0.000 | 23/1 | 60 |
| IP102 (96 cl.) | D12 − B12 | calibrated | +5.6 | [+2.3, +9.0] | 0.002 | 0.004 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | calibrated | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | calibrated | +10.0 | [+1.7, +18.3] | 0.032 | 0.032 | 0.263 | 13/7 | 60 |
| PlantDoc | D12 − B12 | calibrated | +31.8 | [+24.2, +40.9] | 0.001 | 0.003 | 0.000 | 22/1 | 66 |
| PlantVillage | D12 − B12 | calibrated | +27.8 | [+18.9, +36.7] | 0.001 | 0.003 | 0.000 | 35/10 | 90 |

Table 17. As Table 16, pooled over all completed class orders (orders crossed in the bootstrap).

| Data set | Contrast | Metric | Δ | 95% CI | $p$ | $p_{\text{Holm}}$ | McNemar | Discordant | $n$ |
|---|---|---|---|---|---|---|---|---|---|
| IP102 (25 cl.) | D12 − B2 | open | +16.7 | [+8.3, +26.1] | 0.001 | 0.003 | 0.000 | 37/7 | 180 |
| IP102 (96 cl.) | D12 − B12 | open | +4.5 | [+1.7, +7.9] | 0.003 | 0.006 | 0.057 | 11/3 | 177 |
| iNatAg-100 | D12 − B12 | open | +9.0 | [+5.6, +12.4] | 0.001 | 0.003 | 0.002 | 20/4 | 177 |
| iNatAg-25 | D12 − B2 | open | +5.6 | [-1.1, +13.3] | 0.152 | 0.152 | 0.229 | 33/23 | 180 |
| PlantDoc | D12 − B12 | open | +27.3 | [+18.2, +36.4] | 0.001 | 0.003 | 0.000 | 21/3 | 66 |
| PlantVillage | D12 − B12 | open | +26.7 | [+17.8, +35.6] | 0.001 | 0.003 | 0.000 | 32/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | forced | +32.2 | [+23.3, +41.7] | 0.001 | 0.003 | 0.000 | 66/8 | 180 |
| IP102 (96 cl.) | D12 − B12 | forced | +5.6 | [+2.3, +9.0] | 0.002 | 0.003 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | forced | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | forced | +10.0 | [+5.0, +15.6] | 0.001 | 0.003 | 0.025 | 38/20 | 180 |
| PlantDoc | D12 − B12 | forced | +33.3 | [+25.8, +40.9] | 0.001 | 0.003 | 0.000 | 23/1 | 66 |
| PlantVillage | D12 − B12 | forced | +30.0 | [+21.1, +38.9] | 0.001 | 0.003 | 0.000 | 35/8 | 90 |
| IP102 (25 cl.) | D12 − B2 | calibrated | +26.7 | [+17.2, +37.8] | 0.001 | 0.003 | 0.000 | 56/8 | 180 |
| IP102 (96 cl.) | D12 − B12 | calibrated | +5.6 | [+2.3, +9.0] | 0.002 | 0.003 | 0.031 | 14/4 | 177 |
| iNatAg-100 | D12 − B12 | calibrated | +10.7 | [+6.8, +14.1] | 0.001 | 0.003 | 0.000 | 23/4 | 177 |
| iNatAg-25 | D12 − B2 | calibrated | +10.0 | [+5.0, +15.6] | 0.001 | 0.003 | 0.025 | 38/20 | 180 |
| PlantDoc | D12 − B12 | calibrated | +31.8 | [+24.2, +40.9] | 0.001 | 0.003 | 0.000 | 22/1 | 66 |
| PlantVillage | D12 − B12 | calibrated | +27.8 | [+18.9, +36.7] | 0.001 | 0.003 | 0.000 | 35/10 | 90 |

## A. Complete numeric record

Every completed combination of cohort, data set, class order, session and arm, including arms that failed, generated from the prediction logs (results/per_session_*.csv). BA: open-world balanced accuracy (%); $BA_f$: forced choice; F1: open-world macro-F1; Old/New; Rej.: held-out answered UNKNOWN; FR: known answered UNKNOWN; AUROC; Qwen calls and seconds per query.

| Cohort | Data set | Ord. | $t$ | Arm | $n_k$ | $n_h$ | BA | $BA_f$ | F1 | Old | New | Rej. | FR | AUROC | Calls | s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Agent-dev | deepweeds | 42 | 0 | B0 | 15 | 10 | 13 | 53 | 19 | – | 13 | 100 | 60 | 0.94 | 1.0 | 1.4 |
| Agent-dev | deepweeds | 42 | 0 | B1 | 15 | 10 | 80 | 87 | 85 | – | 80 | 40 | 13 | 0.80 | 1.0 | 2.9 |
| Agent-dev | deepweeds | 42 | 0 | B12 | 15 | 10 | 60 | 60 | 61 | – | 60 | 30 | 13 | 0.77 | 1.0 | 3.2 |
| Agent-dev | deepweeds | 42 | 0 | B13 | 15 | 10 | 40 | 67 | 50 | – | 40 | 70 | 60 | 0.59 | 3.0 | 5.8 |
| Agent-dev | deepweeds | 42 | 0 | B2 | 15 | 10 | 60 | 60 | 61 | – | 60 | 30 | 13 | 0.77 | 1.0 | 3.1 |
| Agent-dev | deepweeds | 42 | 0 | B2null | 15 | 10 | 0 | 33 | 0 | – | 0 | 100 | 100 | 0.00 | 1.0 | 3.1 |
| Agent-dev | deepweeds | 42 | 0 | B2perm | 15 | 10 | 20 | 40 | 17 | – | 20 | 80 | 27 | 0.75 | 1.0 | 3.1 |
| Agent-dev | deepweeds | 42 | 0 | B2rev | 15 | 10 | 60 | 67 | 61 | – | 60 | 30 | 13 | 0.77 | 1.0 | 3.1 |
| Agent-dev | deepweeds | 42 | 0 | B2x3 | 8 | 0 | 40 | 50 | 44 | – | 40 | – | 12 | – | 3.0 | 9.4 |
| Agent-dev | deepweeds | 42 | 0 | B3 | 15 | 10 | 33 | 60 | 36 | – | 33 | 50 | 33 | 0.65 | 2.0 | 4.6 |
| Agent-dev | deepweeds | 42 | 0 | B4 | 7 | 0 | 30 | 40 | 38 | – | 30 | – | 29 | – | 4.0 | 9.4 |
| Agent-dev | deepweeds | 42 | 1 | B0 | 25 | 10 | 28 | 36 | 30 | 33 | 20 | 80 | 16 | 0.90 | 1.0 | 1.4 |
| Agent-dev | deepweeds | 42 | 1 | B1 | 25 | 10 | 24 | 28 | 21 | 40 | 0 | 30 | 8 | 0.70 | 1.0 | 4.1 |
| Agent-dev | deepweeds | 42 | 1 | B12 | 25 | 10 | 0 | 36 | 0 | 0 | 0 | 90 | 80 | 0.64 | 1.0 | 2.2 |
| Agent-dev | deepweeds | 42 | 1 | B13 | 25 | 10 | 8 | 40 | 13 | 13 | 0 | 90 | 88 | 0.58 | 3.0 | 5.0 |
| Agent-dev | deepweeds | 42 | 1 | B2 | 25 | 10 | 32 | 36 | 28 | 53 | 0 | 70 | 36 | 0.63 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 1 | B2null | 25 | 10 | 0 | 20 | 0 | 0 | 0 | 100 | 100 | 0.00 | 1.0 | 2.9 |
| Agent-dev | deepweeds | 42 | 1 | B2perm | 25 | 10 | 16 | 24 | 14 | 13 | 20 | 60 | 32 | 0.65 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 1 | B2rev | 25 | 10 | 24 | 28 | 21 | 20 | 30 | 40 | 20 | 0.63 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 1 | B3 | 25 | 10 | 20 | 44 | 20 | 33 | 0 | 80 | 60 | 0.69 | 2.0 | 3.9 |
| Agent-dev | deepweeds | 42 | 2 | B0 | 30 | 10 | 17 | 27 | 21 | 20 | 0 | 50 | 17 | 0.85 | 1.0 | 1.4 |
| Agent-dev | deepweeds | 42 | 2 | B1 | 30 | 10 | 33 | 33 | 24 | 40 | 0 | 10 | 10 | 0.60 | 1.0 | 4.7 |
| Agent-dev | deepweeds | 42 | 2 | B12 | 30 | 10 | 10 | 40 | 14 | 12 | 0 | 100 | 83 | 0.65 | 1.0 | 2.3 |
| Agent-dev | deepweeds | 42 | 2 | B13 | 30 | 10 | 17 | 37 | 20 | 8 | 60 | 90 | 70 | 0.65 | 3.0 | 5.0 |
| Agent-dev | deepweeds | 42 | 2 | B2 | 30 | 10 | 30 | 43 | 30 | 28 | 40 | 60 | 37 | 0.73 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 2 | B2null | 30 | 10 | 0 | 17 | 0 | 0 | 0 | 100 | 100 | 0.00 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 2 | B2perm | 30 | 10 | 13 | 23 | 12 | 16 | 0 | 80 | 50 | 0.73 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 2 | B2rev | 30 | 10 | 30 | 40 | 34 | 28 | 40 | 70 | 50 | 0.74 | 1.0 | 3.0 |
| Agent-dev | deepweeds | 42 | 2 | B3 | 30 | 10 | 30 | 37 | 31 | 28 | 40 | 60 | 53 | 0.57 | 2.0 | 3.8 |
| Agent-dev | mango | 42 | 0 | B0 | 15 | 9 | 33 | 60 | 33 | – | 33 | 100 | 60 | 0.70 | 1.0 | 1.5 |
| Agent-dev | mango | 42 | 0 | B1 | 15 | 9 | 60 | 80 | 69 | – | 60 | 44 | 40 | 0.64 | 1.0 | 3.2 |
| Agent-dev | mango | 42 | 0 | B2 | 15 | 9 | 80 | 87 | 83 | – | 80 | 0 | 13 | 0.53 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 0 | B2null | 15 | 9 | 0 | 33 | 0 | – | 0 | 100 | 100 | 0.00 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 0 | B2perm | 15 | 9 | 13 | 27 | 11 | – | 13 | 11 | 20 | 0.64 | 1.0 | 5.2 |
| Agent-dev | mango | 42 | 0 | B2rev | 15 | 9 | 80 | 87 | 83 | – | 80 | 0 | 13 | 0.58 | 1.0 | 5.2 |
| Agent-dev | mango | 42 | 0 | B3 | 15 | 9 | 67 | 87 | 75 | – | 67 | 0 | 27 | 0.56 | 2.0 | 6.7 |
| Agent-dev | mango | 42 | 0 | C1 | 1 | 0 | 0 | 0 | 0 | – | 0 | – | 100 | – | 4.0 | 9.1 |
| Agent-dev | mango | 42 | 1 | B0 | 25 | 9 | 36 | 52 | 31 | 33 | 40 | 78 | 32 | 0.71 | 1.0 | 1.5 |
| Agent-dev | mango | 42 | 1 | B1 | 25 | 9 | 64 | 68 | 68 | 73 | 50 | 11 | 16 | 0.47 | 1.0 | 4.4 |
| Agent-dev | mango | 42 | 1 | B2 | 25 | 9 | 52 | 56 | 53 | 60 | 40 | 0 | 8 | 0.62 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 1 | B2null | 25 | 9 | 0 | 20 | 0 | 0 | 0 | 100 | 100 | 0.00 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 1 | B2perm | 25 | 9 | 44 | 44 | 35 | 53 | 30 | 0 | 4 | 0.68 | 1.0 | 4.6 |
| Agent-dev | mango | 42 | 1 | B2rev | 25 | 9 | 44 | 48 | 45 | 47 | 40 | 0 | 12 | 0.57 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 1 | B3 | 25 | 9 | 52 | 72 | 54 | 73 | 20 | 44 | 28 | 0.54 | 2.0 | 5.9 |
| Agent-dev | mango | 42 | 2 | B0 | 30 | 9 | 40 | 40 | 32 | 28 | 100 | 78 | 10 | 0.80 | 1.0 | 1.5 |
| Agent-dev | mango | 42 | 2 | B1 | 30 | 9 | 67 | 70 | 59 | 60 | 100 | 11 | 3 | 0.79 | 1.0 | 5.0 |
| Agent-dev | mango | 42 | 2 | B2 | 30 | 9 | 60 | 60 | 52 | 52 | 100 | 22 | 0 | 0.85 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 2 | B2null | 30 | 9 | 0 | 17 | 0 | 0 | 0 | 100 | 100 | 0.00 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 2 | B2perm | 30 | 9 | 40 | 43 | 35 | 44 | 20 | 0 | 3 | 0.75 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 2 | B2rev | 30 | 9 | 60 | 63 | 54 | 52 | 100 | 22 | 3 | 0.83 | 1.0 | 5.1 |
| Agent-dev | mango | 42 | 2 | B3 | 30 | 9 | 60 | 63 | 56 | 52 | 100 | 11 | 7 | 0.63 | 2.0 | 5.3 |
| Agent-fresh | deepweeds | 42 | 0 | B0 | 9 | 6 | 0 | 56 | 0 | – | 0 | 83 | 78 | 0.63 | 1.0 | 2.1 |
| Agent-fresh | deepweeds | 42 | 0 | B1 | 9 | 6 | 56 | 67 | 59 | – | 56 | 33 | 22 | 0.57 | 1.0 | 2.9 |
| Agent-fresh | deepweeds | 42 | 0 | B12 | 9 | 6 | 56 | 78 | 62 | – | 56 | 17 | 33 | 0.59 | 1.0 | 3.2 |
| Agent-fresh | deepweeds | 42 | 0 | B2 | 9 | 6 | 56 | 78 | 62 | – | 56 | 17 | 33 | 0.59 | 1.0 | 3.2 |
| Agent-fresh | deepweeds | 42 | 0 | C1 | 9 | 6 | 0 | 56 | 0 | – | 0 | 83 | 89 | 0.35 | 4.0 | 7.3 |
| Agent-fresh | deepweeds | 42 | 1 | B0 | 15 | 6 | 27 | 40 | 23 | 33 | 17 | 67 | 20 | 0.70 | 1.0 | 2.1 |
| Agent-fresh | deepweeds | 42 | 1 | B1 | 15 | 6 | 33 | 40 | 30 | 22 | 50 | 17 | 33 | 0.72 | 1.0 | 4.1 |
| Agent-fresh | deepweeds | 42 | 1 | B12 | 15 | 6 | 7 | 33 | 8 | 11 | 0 | 100 | 87 | 0.63 | 1.0 | 2.2 |
| Agent-fresh | deepweeds | 42 | 1 | B2 | 15 | 6 | 13 | 27 | 10 | 22 | 0 | 50 | 60 | 0.71 | 1.0 | 3.0 |
| Agent-fresh | deepweeds | 42 | 1 | C1 | 15 | 6 | 13 | 27 | 13 | 22 | 0 | 100 | 80 | 0.62 | 6.0 | 9.8 |
| Agent-fresh | deepweeds | 42 | 2 | B0 | 18 | 6 | 28 | 33 | 31 | 27 | 33 | 33 | 22 | 0.61 | 1.0 | 1.5 |
| Agent-fresh | deepweeds | 42 | 2 | B1 | 18 | 6 | 33 | 39 | 28 | 40 | 0 | 17 | 17 | 0.58 | 1.0 | 4.7 |
| Agent-fresh | deepweeds | 42 | 2 | B12 | 18 | 6 | 6 | 39 | 8 | 7 | 0 | 100 | 89 | 0.69 | 1.0 | 2.3 |
| Agent-fresh | deepweeds | 42 | 2 | B2 | 18 | 6 | 17 | 39 | 15 | 20 | 0 | 33 | 61 | 0.59 | 1.0 | 3.0 |
| Agent-fresh | deepweeds | 42 | 2 | C1 | 18 | 6 | 17 | 17 | 16 | 13 | 33 | 67 | 61 | 0.62 | 7.0 | 11.3 |
| Agent-fresh | insects | 42 | 0 | B0 | 15 | 12 | 73 | 73 | 71 | – | 73 | 58 | 0 | 0.92 | 1.0 | 2.4 |
| Agent-fresh | insects | 42 | 0 | B1 | 15 | 12 | 60 | 60 | 56 | – | 60 | 42 | 0 | 0.93 | 1.0 | 4.4 |
| Agent-fresh | insects | 42 | 0 | B12 | 15 | 12 | 80 | 80 | 78 | – | 80 | 92 | 0 | 0.99 | 1.0 | 3.5 |

| Cohort | Data set | Ord. | $t$ | Arm | $n_k$ | $n_h$ | BA | $\mathrm{BA}_f$ | F1 | Old | New | Rej. | FR | AUROC | Calls | s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Agent-fresh | insects | 42 | 0 | B2 | 15 | 12 | 73 | 73 | 70 | – | 73 | 58 | 0 | 0.93 | 1.0 | 4.9 |
| Agent-fresh | insects | 42 | 0 | C1 | 15 | 12 | 80 | 87 | 83 | – | 80 | 25 | 7 | 0.81 | 6.0 | 12.5 |
| Agent-fresh | insects | 42 | 1 | B0 | 24 | 12 | 79 | 79 | 77 | 80 | 78 | 17 | 0 | 0.91 | 1.0 | 2.2 |
| Agent-fresh | insects | 42 | 1 | B1 | 24 | 12 | 67 | 67 | 65 | 60 | 78 | 8 | 4 | 0.88 | 1.0 | 6.7 |
| Agent-fresh | insects | 42 | 1 | B12 | 24 | 12 | 50 | 58 | 51 | 33 | 78 | 100 | 38 | 0.86 | 1.0 | 2.8 |
| Agent-fresh | insects | 42 | 1 | B2 | 24 | 12 | 67 | 67 | 67 | 60 | 78 | 50 | 12 | 0.86 | 1.0 | 5.1 |
| Agent-fresh | insects | 42 | 1 | C1 | 24 | 12 | 83 | 83 | 84 | 80 | 89 | 25 | 4 | 0.76 | 9.0 | 17.4 |
| Agent-fresh | insects | 42 | 2 | B0 | 33 | 12 | 67 | 67 | 59 | 75 | 44 | 33 | 0 | 0.85 | 1.0 | 1.8 |
| Agent-fresh | insects | 42 | 2 | B1 | 33 | 12 | 55 | 58 | 55 | 62 | 33 | 17 | 9 | 0.84 | 1.0 | 8.3 |
| Agent-fresh | insects | 42 | 2 | B12 | 33 | 12 | 33 | 39 | 32 | 37 | 22 | 100 | 39 | 0.81 | 1.0 | 2.5 |
| Agent-fresh | insects | 42 | 2 | B2 | 33 | 12 | 61 | 64 | 59 | 67 | 44 | 50 | 9 | 0.85 | 1.0 | 5.4 |
| Agent-fresh | insects | 42 | 2 | C1 | 33 | 12 | 67 | 70 | 67 | 79 | 33 | 42 | 12 | 0.71 | 12.0 | 22.0 |